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Microelectronic devices and related memory devices

Abstract

A microelectronic device comprises a stack structure, a stadium structure within the stack structure, and conductive contact structures. The stack structure comprises a vertically alternating sequence of conductive structures and insulative structures arranged in tiers. Each of the tiers comprises one of the conductive structures and one of the insulative structures. The stadium structure comprises a forward staircase structure having steps comprising edges of the tiers, and a reverse staircase structure opposing the forward staircase structure and having additional steps comprising additional edges of the tiers. The conductive contact structures vertically extend to upper vertical boundaries of at least some of the conductive structures of the stack structure at the steps of the forward staircase structure and the additional steps of the reverse staircase structure, and are each integral and continuous with one of the conductive structures. Memory devices, electronic systems, and methods of forming microelectronic devices are also described.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/050,438, filed Oct. 27, 2022, now U.S. Pat. No. 11,942,422, issued Mar. 26, 2024, which is a divisional of U.S. patent application Ser. No. 16/864,823, filed May 1, 2020, now U.S. Pat. No. 11,495,530, issued Nov. 8, 2022, the disclosure of each of which is hereby incorporated herein in its entirety by this reference.

TECHNICAL FIELD

(1) The disclosure, in various embodiments, relates generally to the field of microelectronic device design and fabrication. More specifically, the disclosure relates to microelectronic devices including stadium structures, and to related methods, memory devices, and electronic systems.

BACKGROUND

(2) A continuing goal of the microelectronics industry has been to increase the memory density (e.g., the number of memory cells per memory die) of memory devices, such as non-volatile memory devices (e.g., NAND Flash memory devices). One way of increasing memory density in non-volatile memory devices is to utilize vertical memory array (also referred to as a “three-dimensional (3D) memory array”) architectures. A conventional vertical memory array includes vertical memory strings extending through openings in one or more stack structures including tiers of conductive structures and dielectric materials. Each vertical memory string may include at least one select device coupled in series to a serial combination of vertically stacked memory cells. Such a configuration permits a greater number of switching devices (e.g., transistors) to be located in a unit of die area (i.e., length and width of active surface consumed) by building the array upwards (e.g., vertically) on a die, as compared to structures with conventional planar (e.g., two-dimensional) arrangements of transistors.

(3) Vertical memory array architectures generally include electrical connections between the conductive structures of the tiers of the stack structure(s) of the memory device and access lines (e.g., word lines) so that the memory cells of the vertical memory array can be uniquely selected for writing, reading, or erasing operations. One method of forming such an electrical connection includes forming so-called “staircase” (or “stair step”) structures at edges (e.g., horizontal ends) of the tiers of the stack structure(s) of the memory device. The staircase structure includes individual “steps” defining contact regions of the conductive structures, upon which conductive contact structures can be positioned to provide electrical access to the conductive structures.

(4) As vertical memory array technology has advanced, additional memory density has been provided by forming vertical memory arrays to include additional tiers of conductive structures, and, hence, additional steps in the staircase structures associated therewith. However, increasing the number of steps of a staircase structure without undesirably increasing the overall width (e.g., horizontal footprint) of the staircase structure can decrease the acceptable margin of error associated with different acts in the process of forming the increased number of steps. A conventional process of forming a staircase structure may include repeated acts of trimming a uniform width of a mask (e.g., photoresist) overlying alternating conductive structures and

insulating structures, etching portions of the insulating structures not covered by a remaining portion of the mask, and then etching portions of the conductive structures not covered by remaining portions of the insulating structures. Each of these repeated acts has an associated margin of error permitting the steps of the staircase structure to be suitably sized and positioned to form contact structures thereon. As the number of repeated acts increases, deviation from a desired step width and/or a desired step position may be compounded because errors in the size and/or position of one structure are transferred to subsequently formed structures later in the process. For a large number of steps in the staircase structure, margins of error to achieve suitably sized and positioned steps may be small, such as less than one percent (1%). Achieving such small margins of error can be very difficult using conventional methods, which may result in improperly positioned contact structures and may undesirably decrease yield (e.g., the number of memory cells that are validly programmable and erasable as a percentage of the total number of memory cells in a given batch). In addition, as feature packing densities have increased and margins for formation error have decreased, conventional configurations have resulted in undesirable defects (e.g., contact punch through) as well as current leaks and short circuits that can diminish desired memory device performance, reliability, and durability.

(5) Accordingly, there remains a need for new microelectronic device (e.g., memory device, such as 3D NAND Flash memory device) configurations facilitating enhanced memory density while alleviating the problems of conventional microelectronic device configurations, as well as for new methods of forming the microelectronic devices and new electronic systems including the new microelectronic device configurations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A through 1L are simplified partial cross-sectional views illustrating a method of forming a microelectronic device, in accordance with embodiments of the disclosure.

(2) FIG. 2 is a simplified top-down view of a microelectronic device, in accordance with embodiments of the disclosure.

(3) FIG. 3 is a simplified top-down view of a microelectronic device, in accordance with additional embodiments of the disclosure.

(4) FIGS. 4A through 4C are simplified partial cross-sectional views illustrating a method of forming a microelectronic device, in accordance with additional embodiments of the disclosure.

(5) FIGS. 5A through 5G are simplified partial cross-sectional views illustrating a method of forming a microelectronic device, in accordance with additional embodiments of the disclosure.

(6) FIG. 6 is a schematic block diagram illustrating an electronic system, in accordance with embodiments of the disclosure.

DETAILED DESCRIPTION

(7) The following description provides specific details, such as material compositions, shapes, and sizes, in order to provide a thorough description of embodiments of the disclosure. However, a person of ordinary skill in the art would understand that the embodiments of the disclosure may be practiced without employing these specific details. Indeed, the embodiments of the disclosure may be practiced in conjunction with conventional microelectronic device fabrication techniques employed in the industry. In addition, the description provided below does not form a complete process flow for manufacturing a microelectronic device (e.g., a memory device, such as 3D NAND Flash memory device). The structures described below do not form a complete microelectronic device. Only those process acts and structures necessary to understand the embodiments of the disclosure are described in detail below. Additional acts to form a complete microelectronic device from the structures may be performed by conventional fabrication

techniques.

(8) Drawings presented herein are for illustrative purposes only, and are not meant to be actual views of any particular material, component, structure, device, or system. Variations from the shapes depicted in the drawings as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments described herein are not to be construed as being limited to the particular shapes or regions as illustrated, but include deviations in shapes that result, for example, from manufacturing. For example, a region illustrated or described as box-shaped may have rough and/or nonlinear features, and a region illustrated or described as round may include some rough and/or linear features. Moreover, sharp angles that are illustrated may be rounded, and vice versa. Thus, the regions illustrated in the figures are schematic in nature, and their shapes are not intended to illustrate the precise shape of a region and do not limit the scope of the present claims. The drawings are not necessarily to scale. Additionally, elements common between figures may retain the same numerical designation.

(9) As used herein, the term “memory device” means and includes microelectronic devices exhibiting memory functionality, but not necessarily limited to memory functionality. Stated another way, and by way of non-limiting example only, the term “memory device” includes not only conventional memory (e.g., conventional volatile memory, such as conventional dynamic random access memory (DRAM); conventional non-volatile memory, such as conventional NAND memory), but also includes an application specific integrated circuit (ASIC) (e.g., a system on a chip (SoC)), a microelectronic device combining logic and memory, and a graphics processing unit (GPU) incorporating memory.

(10) As used herein, the terms “vertical,” “longitudinal,” “horizontal,” and “lateral” are in reference to a major plane of a structure and are not necessarily defined by earth's gravitational field. A “horizontal” or “lateral” direction is a direction that is substantially parallel to the major plane of the structure, while a “vertical” or “longitudinal” direction is a direction that is substantially perpendicular to the major plane of the structure. The major plane of the structure is defined by a surface of the structure having a relatively large area compared to other surfaces of the structure.

(11) As used herein, features (e.g., regions, structures, devices) described as “neighboring” one another means and includes features of the disclosed identity (or identities) that are located most proximate (e.g., closest to) one another. Additional features (e.g., additional regions, additional structures, additional devices) not matching the disclosed identity (or identities) of the “neighboring” features may be disposed between the “neighboring” features. Put another way, the “neighboring” features may be positioned directly adjacent one another, such that no other feature intervenes between the “neighboring” features; or the “neighboring” features may be positioned indirectly adjacent one another, such that at least one feature having an identity other than that associated with at least one the “neighboring” features is positioned between the “neighboring” features. Accordingly, features described as “vertically neighboring” one another means and includes features of the disclosed identity (or identities) that are located most vertically proximate (e.g., vertically closest to) one another. Moreover, features described as “horizontally neighboring” one another means and includes features of the disclosed identity (or identities) that are located most horizontally proximate (e.g., horizontally closest to) one another.

(12) As used herein, the terms “comprising,” “including,” “having,” and grammatical equivalents thereof are inclusive or open-ended terms that do not exclude additional, unrecited elements or method steps, but also include the more restrictive terms “consisting of” and “consisting essentially of” and grammatical equivalents thereof. As used herein, the term “may” with respect to a material, structure, feature, or method act indicates that such is contemplated for use in implementation of an embodiment of the disclosure and such term is used in preference to the more restrictive term “is” so as to avoid any implication that other, compatible materials, structures, features, and methods usable in combination therewith should or must be excluded.

(13) As used herein, spatially relative terms, such as “beneath,” “below,” “lower,” “bottom,”

“above,” “upper,” “top,” “front,” “rear,” “left,” “right,” and the like, may be used for case of description to describe one element's or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Unless otherwise specified, the spatially relative terms are intended to encompass different orientations of the materials in addition to the orientation depicted in the figures. For example, if materials in the figures are inverted, elements described as “below” or “beneath” or “under” or “on bottom of” other elements or features would then be oriented “above” or “on top of” the other elements or features. Thus, the term “below” can encompass both an orientation of above and below, depending on the context in which the term is used, which will be evident to one of ordinary skill in the art. The materials may be otherwise oriented (e.g., rotated 90 degrees, inverted, flipped) and the spatially relative descriptors used herein interpreted accordingly. (14) As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

(15) As used herein, “and/or” includes any and all combinations of one or more of the associated listed items.

(16) As used herein, the term “configured” refers to a size, shape, material composition, orientation, and arrangement of one or more of at least one structure and at least one apparatus facilitating operation of one or more of the structure and the apparatus in a pre-determined way.

(17) As used herein, the term “substantially” in reference to a given parameter, property, or condition means and includes to a degree that one of ordinary skill in the art would understand that the given parameter, property, or condition is met with a degree of variance, such as within acceptable tolerances. By way of example, depending on the particular parameter, property, or condition that is substantially met, the parameter, property, or condition may be at least 90.0 percent met, at least 95.0 percent met, at least 99.0 percent met, at least 99.9 percent met, or even 100.0 percent met.

(18) As used herein, “about” or “approximately” in reference to a numerical value for a particular parameter is inclusive of the numerical value and a degree of variance from the numerical value that one of ordinary skill in the art would understand is within acceptable tolerances for the particular parameter. For example, “about” or “approximately” in reference to a numerical value may include additional numerical values within a range of from 90.0 percent to 110.0 percent of the numerical value, such as within a range of from 95.0 percent to 105.0 percent of the numerical value, within a range of from 97.5 percent to 102.5 percent of the numerical value, within a range of from 99.0 percent to 101.0 percent of the numerical value, within a range of from 99.5 percent to 100.5 percent of the numerical value, or within a range of from 99.9 percent to 100.1 percent of the numerical value.

(19) Unless the context indicates otherwise, the materials described herein may be formed by any suitable process including, but not limited to, spin coating, blanket coating, chemical vapor deposition (“CVD”), atomic layer deposition (“ALD”), plasma enhanced ALD, physical vapor deposition (“PVD”) (including sputtering, evaporation, ionized PVD, and/or plasma-enhanced CVD), or epitaxial growth. Depending on the specific material to be formed, the technique for depositing or growing the material may be selected by a person of ordinary skill in the art. In addition, unless the context indicates otherwise, the removal of materials described herein may be accomplished by any suitable process including, but not limited to, etching (e.g., dry etching, wet etching, vapor etching), ion milling, abrasive planarization, or other known methods.

(20) FIGS. 1A through 1L are simplified partial cross-sectional views illustrating embodiments of a method of forming a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device). With the description provided below, it will be readily apparent to one of ordinary skill in the art that the methods and structures described herein may be used in various devices and electronic systems.

(21) Referring to FIG. 1A, a microelectronic device structure **100** may be formed to include a preliminary stack structure **102**, and a patterned mask structure **110** on or over the preliminary stack

structure **102**. The preliminary stack structure **102** includes a vertically alternating (e.g., in the Z-direction) sequence of first insulative structures **104** and second insulative structures **106** arranged in tiers **108**. Each of the tiers **108** of the preliminary stack structure **102** may include at least one of the first insulative structures **104** vertically neighboring at least one of the second insulative structures **106**. The preliminary stack structure **102** may be formed to include any desired number of the tiers **108**, such as greater than or equal to sixteen (16) of the tiers **108**, greater than or equal to thirty-two (32) of the tiers **108**, greater than or equal to sixty-four (64) of the tiers **108**, greater than or equal to one hundred twenty-eight (128) of the tiers **108**, or greater than or equal to two hundred fifty-six (256) of the tiers **108**.

(22) The first insulative structures **104** of the tiers **108** of the preliminary stack structure **102** may be formed of and include at least one electrically insulative material, such one or more of at least one dielectric oxide material (e.g., one or more of a silicon oxide (SiO.sub.x), phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, an aluminum oxide (AlO.sub.x), a hafnium oxide (HfO.sub.x), a niobium oxide (NbO.sub.x), a titanium oxide (TiO.sub.x), a zirconium oxide (ZrO.sub.x), a tantalum oxide (TaO.sub.x), and a magnesium oxide (MgO.sub.x)), at least one dielectric nitride material (e.g., a silicon nitride (SiN.sub.y)), at least one dielectric oxynitride material (e.g., a silicon oxynitride (SiO.sub.xN.sub.y)), and at least one dielectric carboxynitride material (e.g., a silicon carboxynitride (SiO.sub.xC.sub.zN.sub.y)). Formulae including one or more of “x,” “y,” and “z” herein (e.g., SiO.sub.x, AlO.sub.x, HfO.sub.x, NbO.sub.x, TiO.sub.x, SiN.sub.y, SiO.sub.xN.sub.y, SiO.sub.xC.sub.zN.sub.y) represent a material that contains an average ratio of “x” atoms of one element, “y” atoms of another element, and “z” atoms of an additional element (if any) for every one atom of another element (e.g., Si, Al, Hf, Nb, Ti). As the formulae are representative of relative atomic ratios and not strict chemical structure, the first insulative structures **104** may comprise one or more stoichiometric compounds and/or one or more non-stoichiometric compounds, and values of “x,” “y,” and “z” (if any) may be integers or may be non-integers. As used herein, the term “non-stoichiometric compound” means and includes a chemical compound with an elemental composition that cannot be represented by a ratio of well-defined natural numbers and is in violation of the law of definite proportions. Each of the first insulative structures **104** may individually include a substantially homogeneous distribution or a substantially heterogeneous distribution of the at least one electrically insulative material. As used herein, the term “homogeneous distribution” means amounts of a material do not vary throughout different portions (e.g., different horizontal portions, different vertical portions) of a structure. Conversely, as used herein, the term “heterogeneous distribution” means amounts of a material vary throughout different portions of a structure. In some embodiments, each of the first insulative structures **104** exhibits a substantially homogeneous distribution of electrically insulative material. In further embodiments, at least one of the first insulative structures **104** exhibits a substantially heterogeneous distribution of at least one electrically insulative material. One or more of the first insulative structures **104** may, for example, be formed of and include a stack (e.g., laminate) of at least two different electrically insulative materials (e.g., at least two different dielectric materials). In some embodiments, each of the first insulative structures **104** is formed of and includes a dielectric nitride material, such as SiN.sub.y (e.g., Si.sub.3N.sub.4). The first insulative structures **104** may each be substantially planar and may each individually exhibit a desired thickness. In addition, each of the first insulative structures **104** may be substantially the same (e.g., exhibit substantially the same material composition, material distribution, size, and shape) as one another, or at least one of the first insulative structures **104** may be different (e.g., exhibit one or more of a different material composition, a different material distribution, a different size, and a different shape) than at least one other of the first insulative structures **104**. In some embodiments, each of the first insulative structures **104** is substantially the same as each other of the first insulative structures **104**.

(23) The second insulative structures **106** of the tiers **108** of the preliminary stack structure **102**

may be formed of and include at least one additional electrically insulative material. Material compositions of the second insulative structures **106** and the first insulative structures **104** may be selected such that the first insulative structures **104** and the second insulative structures **106** may be selectively removed relative to one another. The second insulative structures **106** may be selectively etchable relative to the first insulative structures **104** during common (e.g., collective, mutual) exposure to a first etchant, and the first insulative structures **104** may be selectively etchable to the second insulative structures **106** during common exposure to a second, different etchant. As used herein, a material is “selectively etchable” relative to another material if the material exhibits an etch rate that is at least about five times (5×) greater than the etch rate of another material, such as about ten times (10×) greater, about twenty times (20×) greater, or about forty times (40×) greater. A material composition of the second insulative structures **106** is different than a material composition of the first insulative structures **104**. The second insulative structures **106** may comprise an additional electrically insulative material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO.sub.x, HfO.sub.x, NbO.sub.x, TiO.sub.x, ZrO.sub.x, TaO.sub.x, and MgO.sub.x), at least one dielectric nitride material (e.g., SiN.sub.y), at least one dielectric oxynitride material (e.g., SiO.sub.xN.sub.y), and at least one dielectric carboxynitride material (e.g., SiO.sub.xC.sub.zN.sub.y). In some embodiments, each of the second insulative structures **106** is formed of and includes a dielectric oxide material, such as SiO.sub.x (e.g., SiO.sub.2). Each of the second insulative structures **106** may individually include a substantially homogeneous distribution of the at least one additional electrically insulative material, or a substantially heterogeneous distribution of the at least one additional electrically insulative material. In some embodiments, each of the second insulative structures **106** of the preliminary stack structure **102** exhibits a substantially homogeneous distribution of additional electrically insulative material. In additional embodiments, at least one of the second insulative structures **106** of the preliminary stack structure **102** exhibits a substantially heterogeneous distribution of at least one additional electrically insulative material. The second insulating structure(s) **106** may, for example, individually be formed of and include a stack (e.g., laminate) of at least two different additional electrically insulative materials. The second insulative structures **106** may each be substantially planar, and may each individually exhibit a desired thickness.

(24) With continued reference to FIG. **1A**, the patterned mask structure **110** may be formed of and include at least one material (e.g., at least one hard mask material) suitable for use as an etch mask to pattern portions of the preliminary stack structure **102** (e.g., portions of the tiers **108**, including portions of the first insulative structures **104** and portions of the second insulative structures **106**) to form apertures (e.g., openings, vias, trenches) vertically extending (e.g., in the Z-direction) into the preliminary stack structure **102**, as described in further detail below. By way of non-limiting example, the patterned mask structure **110** may be formed of and include one or more hard mask materials having etch selectivity relative to the materials of the preliminary stack structure **102** (including the first insulative structures **104** and the second insulative structures **106**). In some embodiments, the patterned mask structure **110** comprises polycrystalline silicon. The patterned mask structure **110** may be homogeneous (e.g., may include only one material layer), or may be heterogeneous (e.g., may include a stack exhibiting at least two different material layers). In addition, the patterned mask structure **110** may exhibit any thickness permitting desired patterning of the preliminary stack structure **102** using patterned mask structure **110**, such as a thickness within a range of from about 1 nanometer (nm) to about 1000 nm.

(25) As shown in FIG. **1A**, the patterned mask structure **110** may include at least one opening **112** (e.g., aperture, via) vertically extending therethrough. The opening **112** may vertically extend (e.g., in the Z-direction) completely through the patterned mask structure **110**, from an upper surface of the patterned mask structure **110** to an upper surface of the preliminary stack structure **102** (e.g., an upper surface of the second insulative structure **106** of an uppermost tier **108** of the preliminary

stack structure **102**).

(26) The patterned mask structure **110** may be formed to exhibit any desired quantity of openings **112**. The quantity of the openings **112** included in the patterned mask structure **110** may at least partially depend on a desired quantity and distribution of trenches to be formed in the preliminary stack structure **102** using the patterned mask structure **110**, which may, in turn, at least partially depend on the quantity of tiers **108** included below the patterned mask structure **110**. In some embodiments, the patterned mask structure **110** is formed to include multiple (e.g., more than one) openings **112**.

(27) A geometric configuration (e.g., shape, dimensions), horizontal position (e.g., in the X-direction, in the Y-direction), and horizontal spacing of each opening **112** in the patterned mask structure **110** at least partially depends on the geometric configuration, horizontal position, and horizontal spacing of the trenches to be subsequently formed in the preliminary stack structure **102** using the patterned mask structure **110**, as described in further detail below. In some embodiments, each opening **112** exhibits substantially the same geometric configuration (e.g., substantially the same shape and substantially the same dimensions) as each other opening **112**, is regularly (e.g., uniformly, non-variably) horizontally spaced apart (e.g., horizontally separated, horizontally distanced) from horizontally neighboring openings **112**, and is substantially horizontally aligned with horizontally neighboring openings **112** in at least one horizontal direction. In additional embodiments, at least one opening **112** in the patterned mask structure **110** exhibits one or more of a different geometric configuration (e.g., a different shape, such as a non-rectangular horizontal cross-sectional shape), different horizontal dimension(s), and/or different horizontal spacing than at least one other opening **112**. For example, at least one opening **112** may exhibit a different (e.g., larger, smaller) width in the X-direction and/or a different length in the Y-direction than at least one other of opening **112**. As another example, some horizontally neighboring openings **112** may be horizontally separated from one another by a first distance, and other horizontally neighboring openings **112** may be horizontally separated from one another by a second distance different than (e.g., less than, greater than) the first distance.

(28) Referring next to FIG. **1B**, portions of at least one upper tier **108** (e.g., a highest tier in the Z-direction, a top tier, an uppermost tier) of the preliminary stack structure **102** exposed by the opening **112** in the patterned mask structure **110** may be removed to expose portions of a relatively lower tier **108** (e.g., a second highest tier in the Z-direction) of the preliminary stack structure **102** vertically underlying (e.g., vertically neighboring) the upper tier **108**. The removal process may partially remove portions of the second insulative structure **106** and the first insulative structure **104** of the upper tier **108** exposed by the opening **112** in the patterned mask structure **110** to uncover (e.g., expose) portions of the second insulative structure **106** of the relatively lower tier **108** vertically underlying the upper tier **108**.

(29) As shown in FIG. **1B**, following the removal process, remaining portions of the upper tier **108** of the preliminary stack structure **102** exposed within the opening **112** may terminate (e.g., end) at or substantially proximate a horizontal centerline **114** (e.g., in the X-direction) of the opening **112** in the patterned mask structure **110**. Thus, newly exposed portions of the relatively lower tier **108** may horizontally extend (e.g., in the X-direction) from and between the horizontal centerline **114** and outer horizontal boundaries (e.g., in the X-direction) of the opening **112** in the patterned mask structure **110**. Put another way, following the removal process, a first width $W_{sub.1}$ of a remaining portion of the upper tier **108** of the preliminary stack structure **102** exposed within the opening **112** may be substantially equal to a second width $W_{sub.2}$ of a newly exposed portion of the relatively lower tier **108** vertically underlying the upper tier **108**. In additional embodiments, the remaining portion of the upper tier **108** of the preliminary stack structure **102** exposed within the opening **112** may terminate at a location horizontally offset from (e.g., in the X-direction) the horizontal centerline **114** (e.g., in the X-direction) of the opening **112** in the patterned mask structure **110**. In such embodiments, the first width $W_{sub.1}$ of the remaining portion of the upper tier **108** of the

preliminary stack structure **102** exposed within the opening **112** may be different than (e.g., greater than, less than) the second width $W_{sub.2}$ of the newly exposed portion of the relatively lower tier **108** vertically underlying the upper tier **108**.

(30) A portion of the upper tier **108** of the preliminary stack structure **102** exposed within the opening **112** in the patterned mask structure **110** may be removed using conventional material removal processes (e.g., conventional photolithographic patterning processes; conventional etching processes, such as a conventional anisotropic etching processes), which are not described in detail herein.

(31) Referring next to FIG. **1C**, the microelectronic device structure **100** may be subjected to one or more material removal processes (e.g., one or more chopping processes) to form at least one trench **116** (e.g., opening, blind via) vertically extending (e.g., in the Z-direction) into the preliminary stack structure **102**. As shown in FIG. **1C**, the trench **116** within the preliminary stack structure **102** may be substantially confined within horizontal boundaries of the opening **112** within the patterned mask structure **110**. In embodiments wherein the patterned mask structure **110** includes multiple openings **112**, different trenches **116** may be formed to extend to different vertical depths than one another within the preliminary stack structure **102**. For example, at least one of the trenches **116** may vertically extend to a relatively lower depth within the preliminary stack structure **102** than at least one other of the trenches **116**. The vertical depths of the trenches **116** relative to one another may at least partially depend on the quantity of tiers **108** of the preliminary stack structure **102**, the quantity of the trenches **116** within the preliminary stack structure **102**, and the horizontal dimensions of the trenches **116**. The trench(es) **116** may be configured to facilitate subsequent formation of vertically extending insulative structures in physical contact with at least some (e.g., each) of the tiers **108** of the preliminary stack structure **102**, as described in further detail below.

(32) As shown in FIG. **1C**, the trench **116** may include a first region **116A** and a second region **116B**. The second region **116B** of the trench **116** may vertically extend (e.g., in the Z-direction) to a relatively lower depth within the preliminary stack structure **102** than the first region **116A** of the trench **116**. For example, the first region **116A** of the trench **116** vertically extend to and terminate at a relatively higher tier **108** of the preliminary stack structure **102**; and the second region **116B** of the trench **116** may vertically extend to and terminate at a relatively lower tier **108** of the preliminary stack structure **102**. The relatively higher tier **108** of the preliminary stack structure **102** and the relatively lower tier **108** of the preliminary stack structure **102** may vertically neighbor one another within the preliminary stack structure **102**. The first region **116A** of the trench **116** may have the first width $W_{sub.1}$ (e.g., in the X-direction) previously described with reference to FIG. **1B**, and the second region **116B** of the trench **116** may have the second width $W_{sub.2}$ (e.g., in the X-direction) previously described with reference to FIG. **1B**. The first region **116A** of the trench **116** may horizontally neighbor the second region **116B** of the trench **116**.

(33) The trench **116** may include a lower vertical boundary **118** (e.g., floor) and horizontal boundaries **120** (e.g., sides). The lower vertical boundary **118** of the trench **116** may have a non-planar topography resulting from the differing vertical depths of the first region **116A** of the trench **116** and the second region **116B** of the trench **116**. For example, a first portion **118A** of the lower vertical boundary **118** of the trench **116** may be defined by the vertical depth of the first region **116A** of the trench **116**, and a second portion **118B** of the lower vertical boundary **118** of the trench **116** may be defined by the vertical depth of the first region **116A** of the trench **116**. The first portion **118A** and the second portion **118B** of the lower vertical boundary **118** of the trench **116** may each be substantially horizontally planar, but may be vertically offset (e.g., in the Z-direction) from one another. The second portion **118B** of the lower vertical boundary **118** of the trench **116** may horizontally neighbor (e.g., in the X-direction) the first portion **118A** of the lower vertical boundary **118** of the trench **116**. In addition, as depicted in FIG. **1C**, the horizontal boundaries **120** of the trench **116** may be substantially vertically planar.

(34) As shown in FIG. **1C**, the trench **116** may be formed to vertically terminate (e.g., vertically

end) at the second insulative structures **106** of two different tiers **108** of the preliminary stack structure **102** (e.g., two vertically neighboring tiers **108** of the preliminary stack structure **102**). For example, the first region **116A** of the trench **116** may vertically terminate at the second insulative structure **106** of a relatively higher tier **108** of the preliminary stack structure **102**; and the second region **116B** of the trench **116** may vertically terminate at the second insulative structure **106** of a relatively lower tier **108** of the preliminary stack structure **102**. Put another way, the first portion **118A** of the lower vertical boundary **118** of the trench **116** may be defined by an exposed portion of the second insulative structure **106** of the relatively higher tier **108** of the preliminary stack structure **102**; and the second portion **118B** of the lower vertical boundary **118** of the trench **116** may be defined by an exposed portion of the second insulative structure **106** of the relatively lower tier **108** of the preliminary stack structure **102**. In additional embodiments, the trench **116** may be formed to vertically terminate at the first insulative structures **104** of two different tiers **108** of the preliminary stack structure **102** (e.g., two vertically neighboring tiers **108** of the preliminary stack structure **102**). For example, the first region **116A** of the trench **116** may vertically terminate at the first insulative structure **104** of a relatively higher tier **108** of the preliminary stack structure **102**; and the second region **116B** of the trench **116** may vertically terminate at the first insulative structure **104** of a relatively lower tier **108** of the preliminary stack structure **102**. Put another way, the first portion **118A** of the lower vertical boundary **118** of the trench **116** may be defined by an exposed portion of the first insulative structure **104** of the relatively higher tier **108** of the preliminary stack structure **102**; and the second portion **118B** of the lower vertical boundary **118** of the trench **116** may be defined by an exposed portion of the first insulative structure **104** of the relatively lower tier **108** of the preliminary stack structure **102**.

(35) Referring next to FIG. **1D**, a first dielectric spacer structure **122-1** may be formed within the opening **112** in the patterned mask structure **110** and the trench **116** in the preliminary stack structure **102**. The first dielectric spacer structure **122-1** may partially (e.g., less than completely) fill each of the opening **112** and the trench **116**, and may be formed on exposed surfaces of the patterned mask structure **110** and the preliminary stack structure **102** within horizontal boundaries (e.g., in the X-direction and the Y-direction) of the trench **116** (and, hence, the opening **112** in the patterned mask structure **110**). For example, as shown in FIG. **1D**, the first dielectric spacer structure **122-1** may be formed directly horizontally adjacent (e.g., horizontally on) side surfaces of the patterned mask structure **110** and the preliminary stack structure **102** at (e.g., coplanar with) the horizontal boundaries **120** of the trench **116**, and may also be formed directly vertically adjacent (e.g., vertically on) upper surfaces of tiers **108** of the preliminary stack structure **102** defining the lower vertical boundary **118** (FIG. **1C**, including the first portion **118A** and the second portion **118B** thereof) of the trench **116**. The first dielectric spacer structure **122-1** may substantially vertically extend across (e.g., in the Z-direction) and cover the side surfaces of the patterned mask structure **110** and the preliminary stack structure **102** at (e.g., coplanar with) the horizontal boundaries **120** of the trench **116**, and may only partially horizontally extend across (e.g., in the X-direction and the Y-direction) and cover the upper surfaces of the tiers **108** of the preliminary stack structure **102** defining the lower vertical boundary **118** (FIG. **1C**) of the trench **116**.

(36) As shown in FIG. **1D**, the first dielectric spacer structure **122-1** may include a first portion **122-1A** and a second portion **122-1B**. The first portion **122-1A** may be formed within horizontal boundaries of the first region **116A** of the trench **116**, and the second dielectric spacer structure **122-2** may be formed within horizontal boundaries of the second region **116B** of the trench **116**. The second portion **122-1B** may vertically extend (e.g., in the Z-direction) to a lower vertical depth than the first portion **122-1A**. The vertical depth of the first portion **122-1A** may correspond to (e.g., be substantially the same as) the vertical depth of the first region **116A** of the trench **116**; and the vertical depth of the second portion **122-1B** may correspond to (e.g., be substantially the same as) the vertical depth of the second region **116B** of the trench **116**. For example, the first portion **122-1A** may vertically extend to and terminate at a relatively higher tier **108** defining the first

portion **118A** (FIG. 1C) of the lower vertical boundary **118** (FIG. 1C) of the trench **116**; and the second portion **122-1B** may vertically extend to and terminate at a relatively lower tier **108** defining the second portion **118B** (FIG. 1C) of the lower vertical boundary **118** (FIG. 1C) of the trench **116**.

(37) The first dielectric spacer structure **122-1** may be formed to exhibit a third width $W_{sub.3}$ (e.g., in the X-direction). The third width $W_{sub.3}$ may at least partially depend on the overall width (e.g., the combination of the first width $W_{sub.1}$ and the second width $W_{sub.2}$ previously described with reference to FIG. 1C) of the trench **116**, and a desired configuration of a stadium structure to be formed within horizontal boundaries of the trench **116**, as described in further detail below. By way of non-limiting example, the third width $W_{sub.3}$ may be within a range of from about 25 nanometers (nm) to about 500 nm.

(38) The first dielectric spacer structure **122-1** may be formed of and include at least one dielectric material having different etch selectivity than the first insulative structures **104** of the preliminary stack structure **102**. The first dielectric spacer structure **122-1** may, for example, have an etch selectively substantially similar to that of the second insulative structures **106** of preliminary stack structure **102**. Portions of the first dielectric spacer structure **122-1** may be employed to protect (e.g., mask) portions of the preliminary stack structure **102** during subsequent processing acts (e.g., subsequent material removal acts, such as subsequent etching acts), as described in further detail below. By way of non-limiting example, the first dielectric spacer structure **122-1** may be formed of and include at least one oxygen-containing dielectric material, such as a one or more of a dielectric oxide material (e.g., one or more of $SiO_{sub.x}$, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, $AlO_{sub.x}$, $HfO_{sub.x}$, $NbO_{sub.x}$, and $TiO_{sub.x}$), a dielectric oxynitride material (e.g., $SiO_{sub.x}N_{sub.y}$), and a dielectric carboxynitride material (e.g., $SiO_{sub.x}C_{sub.z}N_{sub.y}$). A material composition of the first dielectric spacer structure **122-1** may be different than a material composition of the first insulative structures **104** of the preliminary stack structure **102**. The material composition of the first dielectric spacer structure **122-1** may be substantially the same as or may be different than a material composition of the second insulative structures **106** of the preliminary stack structure **102**. In some embodiments, the first dielectric spacer structure **122-1** is formed of and includes $SiO_{sub.x}$ (e.g., $SiO_{sub.2}$).

(39) The first dielectric spacer structure **122-1** may be formed by conformally forming (e.g., through one or more of a conventional ALD process, and a conventional conformal CVD process) a dielectric material (e.g., a dielectric oxide material) on or over surfaces of the preliminary stack structure **102** and the patterned mask structure **110** inside and outside of the trench **116**. Thereafter, the microelectronic device structure **100** may be subjected to at least one material removal (e.g., etching, such as anisotropic dry etching) process to form the first dielectric spacer structure **122-1** (including the first portion **122-1A** and the second portion **122-1B** thereof). The material removal process may substantially (e.g., completely) remove portions of the dielectric material on or over upper surfaces of the patterned mask structure **110** outside of the trench **116**, and may partially remove portions of dielectric material on or over surfaces of the preliminary stack structure **102** at the lower vertical boundary **118** (FIG. 1C) of the trench **116**. As shown in FIG. 1D, the material removal process may also remove portions of the second insulative structures **106** of tiers **108** of the preliminary stack structure **102** defining the lower vertical boundary **118** (FIG. 1C) of the trench **116** to expose (e.g., uncover) portions of the first insulative structures **104** of the tiers **108** within horizontal boundaries of the trench **116**. The exposed portions of the first insulative structures **104** of the tiers **108** may be horizontally bounded by the first dielectric spacer structure **122-1** within the horizontal boundaries of the trench **116**.

(40) Referring next to FIG. 1E, a first dielectric liner structure **124-1** may be formed within remaining (e.g., unfilled) portions of the opening **112** in the patterned mask structure **110** and the trench **116** in the preliminary stack structure **102**. The first dielectric liner structure **124-1** may partially (e.g., less than completely) fill the remaining portions of each of the opening **112** and the

trench **116**, and may be formed on exposed surfaces of the first dielectric spacer structure **122-1** and the preliminary stack structure **102** within horizontal boundaries (e.g., in the X-direction and the Y-direction) of the trench **116** (and, hence, the opening **112** in the patterned mask structure **110**). For example, the first dielectric liner structure **124-1** may be formed directly horizontally adjacent (e.g., horizontally on) inner side surfaces of the first dielectric spacer structure **122-1** (including the first portion **122-1A** and the second portion **122-1B** thereof), and may also be formed directly vertically adjacent (e.g., vertically on) upper surfaces of first insulative structures **104** of tiers **108** of the preliminary stack structure **102** at a modified lower vertical boundary of the trench **116** formed in the processing stage previously described with reference to FIG. **1D**. The first dielectric liner structure **124-1** may substantially vertically extend across (e.g., in the Z-direction) and cover the inner side surfaces of the first dielectric spacer structure **122-1**.

(41) As shown in FIG. **1E**, the first dielectric liner structure **124-1** may include a first portion **124-1A** and a second portion **124-1B**. The first portion **124-1A** may be formed within horizontal boundaries of the first region **116A** of the trench **116**, and the second dielectric liner structure **124-2** may be formed within horizontal boundaries of the second region **116B** of the trench **116**. The first portion **124-1A** may be formed directly horizontally adjacent (e.g., horizontally on) the first portion **122-1A** of the first dielectric spacer structure **122-1**, and the second portion **124-1B** may be formed directly horizontally adjacent (e.g., horizontally on) the second portion **122-1B** of the first dielectric spacer structure **122-1**. The second portion **124-1B** of the first dielectric liner structure **124-1** may vertically extend (e.g., in the Z-direction) to a vertical depth within the preliminary stack structure **102** relatively lower than a vertical depth of the first portion **124-1A** of the first dielectric liner structure **124-1**. The vertical depth of the first portion **124-1A** of the first dielectric liner structure **124-1** may be greater than or equal to the vertical depth of the first portion **122-1A** of the first dielectric spacer structure **122-1**; and the vertical depth of the second portion **124-1B** of the first dielectric liner structure **124-1** may be greater than or equal to the vertical depth of the second portion **122-1B** of the first dielectric spacer structure **122-1**. For example, the first portion **124-1A** of the first dielectric liner structure **124-1** may vertically extend to and terminate at the first insulative structure **104** of a tier **108** of the preliminary stack structure **102** vertically neighboring a lower vertical boundary of the first portion **122-1A** of the first dielectric spacer structure **122-1**; and the second portion **124-1B** of the first dielectric liner structure **124-1** may vertically extend to and terminate at the first insulative structure **104** of another, relatively lower tier **108** of the preliminary stack structure **102** vertically neighboring a lower vertical boundary of the second portion **122-1B** of the first dielectric spacer structure **122-1**.

(42) The first dielectric liner structure **124-1** may be formed to exhibit a fourth width $W_{sub.4}$ (e.g., in the X-direction). The fourth width $W_{sub.4}$ may, for example, be substantially equal to a vertical height (e.g., in the Z-direction) of each of the first insulative structures **104** of the preliminary stack structure **102**. In additional embodiments, the fourth width $W_{sub.4}$ is different than (e.g., less than, greater than) a vertical height of one or more (e.g., each) of the first insulative structures **104** of the preliminary stack structure **102**.

(43) The first dielectric liner structure **124-1** may be formed of and include at least one dielectric material having different etch selectivity than the second insulative structures **106** of the preliminary stack structure **102**. The first dielectric liner structure **124-1** may, for example, have an etch selectively substantially similar to that of the first insulative structures **104** of preliminary stack structure **102**. By way of non-limiting example, if formed, the first dielectric liner structure **124-1** may be formed of and include at least one dielectric nitride material (e.g., $SiN_{sub.y}$). A material composition of the first dielectric liner structure **124-1** may be different than material composition(s) of the first dielectric spacer structure **122-1** (FIG. **1D**) and the second insulative structures **106** of the preliminary stack structure **102**. The material composition of the first dielectric liner structure **124-1** may be substantially the same as or may be different than a material composition of the first insulative structures **104** of the preliminary stack structure **102**. In some

embodiments, the first dielectric liner structure **124-1** is formed of and includes SiN.sub.y (e.g., Si.sub.3N.sub.4).

(44) The first dielectric liner structure **124-1** may be formed by conformally forming (e.g., through one or more of a conventional ALD process, and a conventional conformal CVD process) an additional dielectric material (e.g., a dielectric nitride material) on or over surfaces of the preliminary stack structure **102**, the first dielectric spacer structure **122-1**, and the patterned mask structure **110** inside and outside of the trench **116**. Thereafter, the microelectronic device structure **100** may be subjected to at least one material removal (e.g., etching, such as anisotropic dry etching) process to form the first dielectric liner structure **124-1** (including the first portion **124-1A** and the second portion **124-1B** thereof). The material removal process may substantially (e.g., completely) remove portions of the additional dielectric material on or over upper surfaces of the patterned mask structure **110** and the first dielectric spacer structure **122-1**, and may partially remove portions of additional dielectric material on or over surfaces of the preliminary stack structure **102** at the modified lower vertical boundary of the trench **116**. As shown in FIG. **1E**, the material removal process may also remove portions of the first insulative structures **104** of tiers **108** of the preliminary stack structure **102** vertically neighboring lower vertical boundaries of the first portion **122-1A** and the second portion **122-1B** of the first dielectric spacer structure **122-1** to expose (e.g., uncover) portions of the second insulative structures **106** of relatively vertically lower tiers **108** within horizontal boundaries of the trench **116**. The exposed portions of the second insulative structures **106** of the relatively vertically lower tiers **108** may be horizontally bounded by the first dielectric liner structure **124-1** within the horizontal boundaries of the trench **116**.

(45) Referring next to FIG. **1F**, the microelectronic device structure **100** may be subjected to at least one additional material removal (e.g., etching, such as anisotropic dry etching) processes to increase vertical depths (e.g., in the Z-direction) of remaining (e.g., unfilled) portions of the trench **116**. The additional material removal process may increase a vertical depth of a remaining portion of the first region **116A** of the trench **116**, and may also increase a vertical depth of a remaining portion of the second region **116B** of the trench **116**. Following the additional material removal process, the remaining portion of the second region **116B** of the trench **116** may vertically extend to a relatively lower depth within the preliminary stack structure **102** than the remaining portion of the first region **116A** of the trench **116**. A magnitude of a vertical offset between a lower vertical boundary of the first region **116A** of the trench **116** and a lower vertical boundary of the second region **116B** of the trench **116** may be substantially maintained at the end of the additional material removal process. For example, the additional material removal process may increase a vertical depth of each of the remaining portion of the first region **116A** of the trench **116** and the remaining portion of the second region **116B** of the trench **116** by a distance (e.g., in the Z-direction) substantially equal to the height (e.g., in the Z-direction) of a single (e.g., only one) tier **108** of the preliminary stack structure **102**. Put another way, the additional material removal process may remove an exposed portion of a single (e.g., only one) tier **108** of the preliminary stack structure **102** vertically neighboring a modified lower vertical boundary of the first region **116A** of the trench **116** at the end of the processing stage depicted in FIG. **1E**; and may also remove an exposed portion of another, single (e.g., only one) tier **108** of the preliminary stack structure **102** vertically neighboring a modified lower vertical boundary of the second region **116B** of the trench **116** at the end of the processing stage depicted in FIG. **1E**.

(46) As shown in FIG. **1F**, following the additional material removal process, remaining (e.g., unfilled) portions the trench **116** (including remaining portions of the first region **116A** and the second region **116B** thereof) may vertically terminate (e.g., vertically end) at the second insulative structures **106** of two different tiers **108** of the preliminary stack structure **102** (e.g., two vertically neighboring tiers **108** of the preliminary stack structure **102**). For example, the remaining portion of the first region **116A** of the trench **116** may vertically terminate at the second insulative structure **106** of a relatively higher tier **108** of the preliminary stack structure **102**; and the remaining portion

of the second region **116B** of the trench **116** may vertically terminate at the second insulative structure **106** of a relatively lower tier **108** of the preliminary stack structure **102**. In additional embodiments, following the additional material removal process remaining (e.g., unfilled) portions the trench **116** may vertically terminate at the first insulative structures **104** of two different tiers **108** of the preliminary stack structure **102** (e.g., two vertically neighboring tiers **108** of the preliminary stack structure **102**). For example, the first region **116A** of the trench **116** may vertically terminate at the first insulative structure **104** of a relatively higher tier **108** of the preliminary stack structure **102**; and the second region **116B** of the trench **116** may vertically terminate at the first insulative structure **104** of a relatively lower tier **108** of the preliminary stack structure **102**.

(47) Referring next to FIG. **1G**, a second dielectric spacer structure **122-2** may be formed within a remaining (e.g., unfilled) portion the opening **112** in the patterned mask structure **110** and in remaining, deepened portions of the trench **116** in the preliminary stack structure **102**. The second dielectric spacer structure **122-2** may partially (e.g., less than completely) fill the remaining portion of the opening **112** and the remaining, deepened portions of the trench **116**; and may be formed on exposed surfaces of the first dielectric liner structure **124-1** (FIG. **1F**) and the preliminary stack structure **102** within horizontal boundaries (e.g., in the X-direction and the Y-direction) of the trench **116** (and, hence, the opening **112** in the patterned mask structure **110**). For example, the second dielectric spacer structure **122-2** may be formed directly horizontally adjacent (e.g., horizontally on) exposed inner side surfaces of the first dielectric liner structure **124-1** (FIG. **1F**, which includes the first portion **124-1A** and the second portion **124-1B** shown in FIG. **1G**) and the preliminary stack structure **102** within the trench **116**, and may also be formed directly vertically adjacent (e.g., vertically on) upper surfaces of second insulative structures **106** of tiers **108** of the preliminary stack structure **102** at a modified lower vertical boundary of the trench **116** formed in the processing stage previously described with reference to FIG. **1G**. The second dielectric spacer structure **122-2** may substantially vertically extend across (e.g., in the Z-direction) and cover the inner side surfaces of the first dielectric liner structure **124-1** (FIG. **1F**).

(48) As shown in FIG. **1G**, the second dielectric spacer structure **122-2** may include a first portion **122-2A** and a second portion **122-2B**. The first portion **122-2A** may be formed within horizontal boundaries of the first region **116A** of the trench **116**, and the second portion **122-2B** may be formed within horizontal boundaries of the second region **116B** of the trench **116**. The first portion **122-2A** of the second dielectric spacer structure **122-2** may be formed directly horizontally adjacent (e.g., horizontally on) the first portion **124-1A** of the first dielectric liner structure **124-1**, and the second portion **122-2B** may be formed directly horizontally adjacent (e.g., horizontally on) the second portion **124-1B** of the first dielectric liner structure **124-1**. The second portion **122-2B** of the second dielectric spacer structure **122-2** may vertically extend (e.g., in the Z-direction) to a lower vertical depth within the preliminary stack structure **102** than the first portion **122-2A** of the second dielectric spacer structure **122-2**. The vertical depth of the first portion **122-2A** of the second dielectric spacer structure **122-2** may be greater than the vertical depth of the first portion **124-1A** of the first dielectric liner structure **124-1**; and the vertical depth of the second portion **122-2B** of the second dielectric spacer structure **122-2** may be greater than the vertical depth of the second portion **124-1B** of the first dielectric liner structure **124-1**. For example, the first portion **122-2A** of the second dielectric spacer structure **122-2** may vertically extend to and terminate at the first insulative structure **104** of a relatively higher tier **108** of the preliminary stack structure **102**; and the second portion **122-2B** of the second dielectric spacer structure **122-2** may vertically extend to and terminate at the first insulative structure **104** of another, relatively lower tier **108** of the preliminary stack structure **102** vertically neighboring the relatively higher tier **108** of the preliminary stack structure **102**.

(49) In some embodiments, the second dielectric spacer structure **122-2** is formed to exhibit a width (e.g., in the X-direction) substantially equal to the third width $W_{sub.3}$ of the first dielectric

spacer structure **122-1** (FIG. **1D**, including the first portion **122-1A** and the second portion **122-1B** shown in FIG. **1G**). In additional embodiments, the second dielectric spacer structure **122-2** is formed to exhibit a width different than (e.g., greater than, less than) the third width $W_{sub.3}$. (50) The second dielectric spacer structure **122-2** may be formed of and include at least one dielectric material having different etch selectivity than the first insulative structures **104** of the preliminary stack structure **102** and the first dielectric liner structure **124-1** (FIG. **1F**). The second dielectric spacer structure **122-2** may, for example, have etch selectivity substantially similar to that of the second insulative structures **106** of preliminary stack structure **102** and the first dielectric spacer structure **122-1** (FIG. **1D**). By way of non-limiting example, the second dielectric spacer structure **122-2** may be formed of and include at least one oxygen-containing dielectric material, such as a one or more of a dielectric oxide material (e.g., one or more of $SiO_{sub.x}$, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, $AlO_{sub.x}$, $HfO_{sub.x}$, $NbO_{sub.x}$, and $TiO_{sub.x}$), a dielectric oxynitride material (e.g., $SiO_{sub.x}N_{sub.y}$), and a dielectric carboxynitride material (e.g., $SiO_{sub.x}C_{sub.z}N_{sub.y}$). A material composition of the second dielectric spacer structure **122-2** may be different than material composition(s) of the first insulative structures **104** of the preliminary stack structure **102** and the first dielectric liner structure **124-1** (FIG. **1F**). The material composition of the second dielectric spacer structure **122-2** may be substantially the same as or may be different than a material composition of the second insulative structures **106** of the preliminary stack structure **102** and the first dielectric spacer structure **122-1** (FIG. **1D**). In some embodiments, the second dielectric spacer structure **122-2** is formed of and includes $SiO_{sub.x}$ (e.g., $SiO_{sub.2}$).

(51) The second dielectric spacer structure **122-2** may be formed by conformally forming (e.g., through one or more of a conventional ALD process, and a conventional conformal CVD process) a dielectric material (e.g., a dielectric oxide material) on or over surfaces of the preliminary stack structure **102**, the first dielectric spacer structure **122-1**, the first dielectric liner structure **124-1**, and the patterned mask structure **110** inside and outside of the trench **116**. Thereafter, the microelectronic device structure **100** may be subjected to at least one material removal (e.g., etching, such as anisotropic dry etching) process to form the second dielectric spacer structure **122-2** (including the first portion **122-2A** and the second portion **122-2B** thereof). The material removal process may substantially (e.g., completely) remove portions of the dielectric material on or over upper surfaces of the patterned mask structure **110**, the first dielectric spacer structure **122-1**, and the first dielectric liner structure **124-1**; and may partially remove portions of the dielectric material on or over surfaces of the preliminary stack structure **102** at a modified lower vertical boundary of the trench **116**. As shown in FIG. **1G**, the material removal process may also remove portions of the second insulative structures **106** of tiers **108** of the preliminary stack structure **102** defining the modified lower vertical boundary of the trench **116** to expose (e.g., uncover) portions of the first insulative structures **104** of the tiers **108** within horizontal boundaries of the trench **116**. The exposed portions of the first insulative structures **104** of the tiers **108** may be horizontally bounded by the second dielectric spacer structure **122-2** within the horizontal boundaries of the trench **116**.

(52) Referring next to FIG. **1H**, following the processing stage described with reference to FIG. **1G**, the microelectronic device structure **100** may be subjected to additional processing acts similar to those previously described with reference to FIGS. **1E** through **1G** to fill remaining portions of the trench **116** (FIG. **1G**) with more of the dielectric liner structures **124** (e.g., in addition to the first dielectric liner structure **124-1**) and more of the dielectric spacer structures **122** (e.g., in addition to the first dielectric spacer structure **122-1** and the second dielectric spacer structure **122-2**), and form a stadium structure **126** vertically underlying and in contact with the dielectric liner structures **124** and the dielectric spacer structures **122**. As shown in FIG. **1H**, the stadium structure **126** may include a forward staircase structure **128** including steps **132** comprising horizontal ends of pairs of the tiers **108** of the preliminary stack structure **102**; a reverse staircase structure **130** including additional steps **134** comprising horizontal ends of additional pairs of the tiers **108** of the

preliminary stack structure **102**; and a central region **136** horizontally interposed between the forward staircase structure **128** and the reverse staircase structure **130**. The central region **136** of the stadium structure **126** may, for example, constitute an intersection region and vertical termination location for the forward staircase structure **128** and the reverse staircase structure **130**. As described in further detail below, different dielectric spacer structures **122** and different dielectric liner structures **124** may be formed to physically contact different steps **132** of the forward staircase structure **128** and different additional steps **134** of the reverse staircase structure **130** than one another.

(53) As shown in FIG. 1H, the dielectric liner structures **124** formed within the horizontal boundaries **120** of the trench **116** (FIG. 1G) may include the first dielectric liner structure **124-1** (comprising the first portion **124-1A** and the second portion **124-1B** thereof); a second dielectric liner structure **124-2** (comprising a first portion **124-2A** and a second portion **124-2B** thereof) vertically extending relatively deeper into the preliminary stack structure **102** than the first dielectric liner structure **124-1**; a third dielectric liner structure **124-3** (comprising a first portion **124-3A** and a second portion **124-3B** thereof) vertically extending relatively deeper into the preliminary stack structure **102** than the second dielectric liner structure **124-2**; and a fourth dielectric liner structure **124-4** (comprising a first portion **124-4A** and a second portion **124-4B** thereof) vertically extending relatively deeper into the preliminary stack structure **102** than the third dielectric liner structure **124-3**. In additional embodiments, a different quantity of the dielectric liner structures **124** is formed within the horizontal boundaries **120** of the trench **116** (FIG. 1G). For example, greater than four (4) (e.g., greater than or equal to ten (10), greater than or equal to twenty (20), greater than or equal to fifty (50)) of the dielectric liner structures **124** may be formed within the horizontal boundaries **120** of the trench **116** (FIG. 1G), or less than four (4) of the dielectric liner structures **124** may be formed within the horizontal boundaries **120** of the trench **116** (FIG. 1G). The quantity of dielectric liner structures **124** formed within the horizontal boundaries of the trench **116** (e.g., through processing acts similar to those previously described with reference to FIG. 1E) may at least partially depend on the horizontal dimensions of the trench **116** (FIG. 1G), the dielectric spacer structures **122**, and the dielectric liner structures **124**.

(54) Still referring to FIG. 1H, the dielectric liner structures **124** may physically contact the first insulative structures **104** of different tiers **108** of the preliminary stack structure **102** at the steps **132** and the additional steps **134** of the stadium structure **126**. As shown in FIG. 1H, the first portions **124-1A**, **124-2A**, **124-3A**, **124-4A** of the different dielectric liner structures **124** (e.g., the first dielectric liner structure **124-1**, the second dielectric liner structure **124-2**, the third dielectric liner structure **124-3**, the fourth dielectric liner structure **124-4**) may physically contact at least some of the additional steps **134** of the reverse staircase structure **130** of the stadium structure **126**; and the second portions **124-1B**, **124-2B**, **124-3B**, **124-4B** of the different dielectric liner structures **124** (e.g., the first dielectric liner structure **124-1**, the second dielectric liner structure **124-2**, the third dielectric liner structure **124-3**, the fourth dielectric liner structure **124-4**) may physically contact at least some of the steps **132** of the forward staircase structure **128** of the stadium structure **126**.

(55) As depicted in FIG. 1H, the dielectric liner structures **124** formed within the horizontal boundaries **120** of the trench **116** (FIG. 1G) may individually be horizontally positioned at horizontal boundaries (e.g., in the X-direction, in the Y-direction) of the steps **132** and the additional steps **134** of the stadium structure **126** that the dielectric liner structures **124** physically contact. Put another way, a horizontal boundary (e.g., an inner horizontal boundary positioned relatively closer to the central region **136** of the stadium structure **126** than another, outer horizontal boundary) of each dielectric liner structure **124** may be substantially coplanar with horizontal boundaries (e.g., the inner horizontal boundaries horizontally positioned relatively closer to the central region **136** of the stadium structure **126** than other, outer horizontal boundaries) of the step **132** and the additional step **134** of the stadium structure **126** that the dielectric liner structure

124 physically contacts.

(56) The dielectric liner structures **124** may horizontally alternate with the dielectric spacer structures **122** within the horizontal boundaries **120** of the trench **116** (FIG. **1G**). By way of non-limiting example, as shown in FIG. **1H**, the first dielectric liner structure **124-1** may be horizontally interposed between the first dielectric spacer structure **122-1** and the second dielectric spacer structure **122-2**; the second dielectric liner structure **124-2** may be horizontally interposed between the second dielectric spacer structure **122-2** and a third dielectric spacer structure **122-3**; the third dielectric liner structure **124-3** may be horizontally interposed between the third dielectric spacer structure **122-3** and a fourth dielectric spacer structure **122-4**; and the fourth dielectric liner structure **124-4** may be horizontally interposed between the fourth dielectric spacer structure **122-4** and a fifth dielectric spacer structure **122-5**. Each dielectric liner structure **124** may horizontally extend from and between the dielectric spacer structures **122** horizontally neighboring the dielectric liner structure **124**.

(57) Each of the dielectric liner structures **124** may exhibit substantially the same width (e.g., in the X-direction) as each other of the dielectric liner structures **124**, or at least one of the dielectric liner structures **124** may exhibit a different width than at least one other of the dielectric liner structures **124**. In some embodiments, the dielectric liner structures **124** each individually exhibit a width substantially equal to a vertical height (e.g., in the Z-direction) of one or more of the first insulative structures **104** of the preliminary stack structure **102** that the dielectric liner structure **124** physically contacts. As a non-limiting example, the dielectric liner structures **124** may each individually exhibit the fourth width $W_{sub.4}$ (FIG. **1G**) previously described with reference to the first dielectric liner structure **124-1**. In additional embodiments, at least one the dielectric liner structure **124** exhibits a width different than (e.g., less than, greater than) a vertical height of one or more of the first insulative structures **104** of the preliminary stack structure **102** that the dielectric liner structure **124** physically contacts.

(58) Still referring to FIG. **1H**, the dielectric spacer structures **122** formed within the horizontal boundaries **120** of the trench **116** (FIG. **1G**) may include the first dielectric spacer structure **122-1** (comprising the first portion **122-1A** and the second portion **122-1B** thereof); the second dielectric spacer structure **122-2** (comprising the first portion **122-2A** and the second portion **122-2B** thereof) vertically extending relatively deeper into the preliminary stack structure **102** than the first dielectric spacer structure **122-1**; the third dielectric spacer structure **122-3** (comprising a first portion **122-3A** and a second portion **122-3B** thereof) vertically extending relatively deeper into the preliminary stack structure **102** than the second dielectric spacer structure **122-2**; the fourth dielectric spacer structure **122-4** (comprising a first portion **122-4A** and a second portion **122-4B** thereof) vertically extending relatively deeper into the preliminary stack structure **102** than the third dielectric spacer structure **122-3**; and the fifth dielectric spacer structure **122-5** vertically extending relatively deeper into the preliminary stack structure **102** than the fourth dielectric spacer structure **122-4**. In additional embodiments, a different quantity of the dielectric spacer structures **122** is formed within the horizontal boundaries **120** of the trench **116** (FIG. **1G**). For example, greater than five (5) (e.g., greater than or equal to ten (10), greater than or equal to twenty (20), greater than or equal to fifty (50)) of the dielectric spacer structures **122** may be formed within the horizontal boundaries **120** of the trench **116** (FIG. **1G**), or less than five (5) of the dielectric spacer structures **122** may be formed within the horizontal boundaries **120** of the trench **116** (FIG. **1G**). The quantity of dielectric spacer structures **122** formed within the horizontal boundaries **120** of the trench **116** (e.g., through processing acts similar to those previously described with reference to FIG. **1E**) may at least partially depend on the horizontal dimensions of the trench **116** (FIG. **1G**), the dielectric spacer structures **122**, and the dielectric liner structures **124**.

(59) The dielectric spacer structures **122** may be formed horizontally adjacent the dielectric spacer structures **122** over different steps **132** and additional steps **134** of the stadium structure **126**. As shown in FIG. **1H**, the first portions **122-1A**, **122-2A**, **122-3A**, **122-4A** of the different dielectric

spacer structures **122** (e.g., the first dielectric spacer structure **122-1**, the second dielectric spacer structure **122-2**, the third dielectric spacer structure **122-3**, the fourth dielectric spacer structure **122-4**) may be formed over and within horizontal boundaries of at least some of the additional steps **134** of the reverse staircase structure **130** of the stadium structure **126**; and the second portions **122-1B**, **122-2B**, **122-3B**, **122-4B** of the different dielectric spacer structures **122** (e.g., the first dielectric spacer structure **122-1**, the second dielectric spacer structure **122-2**, the third dielectric spacer structure **122-3**, the fourth dielectric spacer structure **122-4**) may be formed over and within horizontal boundaries of at least some of the steps **132** of the forward staircase structure **128** of the stadium structure **126**. The fifth dielectric spacer structure **122-5** may be formed over and within horizontal boundaries of the central region **136** of the stadium structure **126**.

(60) As previously described, the dielectric spacer structures **122** may horizontally alternate with the dielectric spacer structures **122** within the horizontal boundaries **120** of the trench **116** (FIG. **1G**). By way of non-limiting example, as shown in FIG. **1H**, the first dielectric spacer structure **122-1** may be horizontally interposed between the first dielectric liner structure **124-1** and portions of the preliminary stack structure **102** and the patterned mask structure **110** at the horizontal boundaries **120** of the trench **116** (FIG. **1G**); the second dielectric spacer structure **122-2** may be horizontally interposed between the first dielectric liner structure **124-1** and the second dielectric liner structure **124-2**; the third dielectric spacer structure **122-3** may be horizontally interposed between the second dielectric liner structure **124-2** and the third dielectric liner structure **124-3**; the fourth dielectric spacer structure **122-4** may be horizontally interposed between the third dielectric liner structure **124-3** and the fourth dielectric liner structure **124-4**; and the fifth dielectric spacer structure **122-5** may be horizontally interposed between the first portion **124-4A** of the fourth dielectric liner structure **124-4** and the second portion **124-4B** of the fourth dielectric liner structure **124-4**.

(61) Each of the dielectric spacer structures **122** may exhibit substantially the same width (e.g., in the X-direction) as each other of the dielectric spacer structures **122**, or at least one of the dielectric spacer structures **122** may exhibit a different width than at least one other of the dielectric spacer structures **122**. In some embodiments, the dielectric spacer structure **122** (e.g., the fifth dielectric spacer structure **122-5**) positioned within horizontal boundaries of the central region **136** of the stadium structure **126** exhibits a different width (e.g., a greater width, a smaller width) than one or more (e.g., each) other of the dielectric spacer structures **122**. In additional embodiments, the dielectric spacer structure **122** (e.g., the fifth dielectric spacer structure **122-5**) positioned within horizontal boundaries of the central region **136** of the stadium structure **126** exhibits substantially the same width (e.g., the third width $W_{sub.3}$ (FIG. **1G**)) as each other of the dielectric spacer structures **122**.

(62) With continued reference to FIG. **1H**, the forward staircase structure **128** and the reverse staircase structure **130** of the stadium structure **126** may horizontally oppose one another (e.g., in the X-direction) and may be partially vertically offset from one another (e.g., in the Z-direction) within the horizontal boundaries **120** of the trench **116** (FIG. **1G**). A phantom line extending from a top (e.g., a vertically uppermost one of the steps **132**) of the forward staircase structure **128** to a bottom (e.g., a vertically lowermost one of the steps **132**) of the forward staircase structure **128** may have a positive slope, and another phantom line extending from a top (e.g., a vertically uppermost one of the additional steps **134**) of the reverse staircase structure **130** to a bottom (e.g., a vertically lowermost one of the additional steps **134**) of the reverse staircase structure **130** may have a negative slope. In addition, upper boundaries (e.g., upper surfaces) of the steps **132** of the forward staircase structure **128** may be vertically offset from upper boundaries (e.g., upper surfaces) of the additional steps **134** of the reverse staircase structure **130**. In some embodiments, upper boundaries of the steps **132** of the forward staircase structure **128** are vertically offset from (e.g., vertically underlie) upper boundaries of the additional steps **134** of the reverse staircase structure **130** most vertically proximate thereto by a distance equal to the thickness of one of the

tiers **108** of the preliminary stack structure **102**.

(63) Individual steps **132** of the forward staircase structure **128** of the stadium structure **126** may comprise horizontal ends of two vertically neighboring tiers **108** of the preliminary stack structure **102**. In addition, individual additional steps **134** of the reverse staircase structure **130** of the stadium structure **126** may also comprise horizontal ends of two vertically neighboring tiers **108** of the preliminary stack structure **102**. Individual steps **132** of the stadium structure **126** vertically neighboring individual additional steps **134** of the stadium structure **126** may share a tier **108** of the preliminary stack structure **102** with the individual additional steps **134**. For example, one of the two vertically neighboring tiers **108** of the preliminary stack structure **102** associated with (e.g., defining) an individual step **132** may also be one of the two vertically neighboring tiers **108** of the preliminary stack structure **102** associated with an individual additional step **134** vertically neighboring the individual step **132**. A tier **108** shared by an individual step **132** and an individual additional step **134** vertically neighboring the individual step **132** may form a vertically upper portion of the individual step **132** and a vertically lower portion of the individual additional step **134**, or vice versa. As a non-limiting example, an uppermost additional step **134** of the reverse staircase structure **130** may comprise horizontal ends of two vertically neighboring tiers **108** of the preliminary stack structure **102**; and an uppermost step **132** of the forward staircase structure **128** may comprise horizontal ends of two additional vertically neighboring tiers **108** of the preliminary stack structure **102**, wherein one of the two additional vertically neighboring tiers **108** is the same tier **108** as one of the two vertically neighboring tiers **108**. A first of the two vertically neighboring tiers **108** may form an upper portion of the uppermost additional step **134** and may physically contact the first portion **124-1A** of the first dielectric liner structure **124-1**; and a second of the two vertically neighboring tiers **108** may form a lower portion of the uppermost additional step **134** and may not physically contact the first portion **124-1A** of the first dielectric liner structure **124-1**. In turn, the tier **108** forming the lower portion of the upper additional step **134** may constitute a first of the two additional vertically neighboring tiers **108** and may form an upper portion of the uppermost step **132** in physical contact with the second portion **124-1B** of the first dielectric liner structure **124-1**; and a second of the two additional vertically neighboring tiers **108** may form a lower portion of the uppermost step **132** and may not physically contact the second portion **124-1B** of the first dielectric liner structure **124-1**.

(64) Referring next to FIG. **1I**, the patterned mask structure **110** (FIG. **1H**) and portions (e.g., upper portions) of the dielectric spacer structures **122** and the dielectric liner structures **124** outside of the vertical boundaries of the trench **116** (FIG. **1C**) may be removed while retaining other portions (e.g., lower portions) of the dielectric spacer structures **122** and the dielectric liner structures **124** within the vertical boundaries of the trench **116** (FIG. **1C**). As shown in FIG. **1I**, the removal process may expose an uppermost surface of the preliminary stack structure **102** of the microelectronic device structure **100**. Uppermost boundaries (e.g., uppermost surfaces) of remaining portions (e.g., lower portions) of the dielectric spacer structures **122** and the dielectric liner structures **124** may be substantially coplanar with the exposed uppermost surface of the preliminary stack structure **102**. Put another way, following the removal process, uppermost boundaries (e.g., uppermost surfaces) of the dielectric spacer structures **122**, the dielectric liner structures **124**, and the preliminary stack structure **102** may form a substantially planar uppermost surface **138** of the microelectronic device structure **100**.

(65) The patterned mask structure **110** (FIG. **1H**) and portions (e.g., upper portions) of the dielectric spacer structures **122** and the dielectric liner structures **124** outside of the vertical boundaries of the trench **116** (FIG. **1C**) may be removed using conventional processes (e.g., conventional etching processes, conventional planarization processes), which are not described in detail herein. For example, the patterned mask structure **110** (FIG. **1H**) and portions (e.g., upper portions) of the dielectric spacer structures **122** and the dielectric liner structures **124** outside of the vertical boundaries of the trench **116** (FIG. **1C**) may be removed using one or more of an etching process

(e.g., a wet etching process) and a chemical-mechanical planarization (CMP) process.

(66) Referring next to FIG. 1J, a capping material **140** may be formed on or over the substantially planar uppermost surface **138** (FIG. 1I) of the microelectronic device structure **100**. The capping material **140** may horizontally extend continuously across the substantially planar uppermost surface **138** (FIG. 1I) of the microelectronic device structure **100**. The capping material **140** may cover (e.g., cap) uppermost surfaces of the dielectric liner structures **124**, and may facilitate replacement of dielectric material(s) of the dielectric liner structures **124** and at least some of the first insulative structures **104** of the preliminary stack structure **102** with one or more conductive materials through so called “replacement gate” or “gate last” processing acts, as described in further detail below.

(67) The capping material **140** may be formed of and include at least one dielectric material having different etch selectivity than the first insulative structures **104** of the preliminary stack structure **102** and the dielectric liner structures **124**. The capping material **140** may, for example, have etch selectively substantially similar to that of the second insulative structures **106** of preliminary stack structure **102** and the dielectric spacer structures **122**. By way of non-limiting example, the capping material **140** may be formed of and include at least one oxygen-containing dielectric material, such as a one or more of a dielectric oxide material (e.g., one or more of SiO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO.sub.x, HfO.sub.x, NbO.sub.x, and TiO.sub.x), a dielectric oxynitride material (e.g., SiO.sub.xN.sub.y), and a dielectric carboxynitride material (e.g., SiO.sub.xC.sub.zN.sub.y). A material composition of the capping material **140** may be different than material composition(s) of the first insulative structures **104** of the preliminary stack structure **102** and the dielectric liner structures **124**. The material composition of the capping material **140** may be substantially the same as or may be different than material composition(s) of the second insulative structures **106** of the preliminary stack structure **102** and the dielectric spacer structures **122**. In some embodiments, the capping material **140** is formed of and includes SiO.sub.x (e.g., SiO.sub.2).

(68) The capping material **140** may be formed using conventional processes (e.g., conventional deposition processes, such as one or more PVD, CVD, ALD, and spin-coating; conventional material removal processes, such as conventional polishing processes). Such processes are known in the art and, therefore, are not described in detail herein. In some embodiments, a dielectric material is conventionally deposited on or over the substantially planar uppermost surface **138** (FIG. 1I) of the microelectronic device structure **100** to form the capping material **140**, and then the capping material **140** is subjected to at least one polishing process (e.g., at least one buffing process) to mitigate (e.g., substantially remove) undesirable irregularities (e.g., deviations from planarity) at an uppermost surface thereof. The uppermost surface of the capping material **140** may be substantially planar.

(69) Referring next to FIG. 1K, the first portions (e.g., the first portions **124-1A**, **124-2A**, **124-3A**, **124-4A** shown in FIG. 1J) of the dielectric liner structures **124** (e.g., the dielectric liner structures **124-1**, **124-2**, **124-3**, **124-4** shown in FIG. 1J) may be physically separated (e.g., physically partitioned) from the second portions (e.g., the second portions **124-1B**, **124-2B**, **124-3B**, **124-4B** shown in FIG. 1J) of the dielectric liner structures **124** (FIG. 1J); and dielectric material(s) of the dielectric liner structures **124** (FIG. 1J) and at least some of the first insulative structures **104** (FIG. 1J) of the preliminary stack structure **102** (FIG. 1J) may be at least partially replaced with conductive material(s) to form conductive structures **142** from the first insulative structures **104** (FIG. 1J), conductive contact structures **144** from the first portions of the dielectric liner structures **124** (FIG. 1J), and additional conductive contact structures **146** from the second portions of the dielectric liner structures **124** (FIG. 1J). As shown in FIG. 1K, the process may modify the tiers **108** (FIG. 1J) of the preliminary stack structure **102** (FIG. 1J) to form a stack structure **148** including a vertically alternating (e.g., in the Z-direction) sequence of the conductive structures **142** and insulative structures **150** arranged in modified tiers **152**. The insulative structures **150** may

correspond to remainders (e.g., remaining portions, unremoved portions) of the second insulative structures **106** (FIG. 1J) of the preliminary stack structure **102** (FIG. 1J). The modified tiers **152** of the stack structure **148** may be formed from and correspond to the tiers **108** (FIG. 1J) of the preliminary stack structure **102** (FIG. 1J). The modified tiers **152** of the stack structure **148** may each individually include a conductive structure **142** vertically neighboring an insulative structure **150**. In embodiments wherein only portions (e.g., less than entireties) of the first insulative structures **104** (FIG. 1J) of the preliminary stack structure **102** (FIG. 1J) are replaced with the conductive structures **142**, the modified tiers **152** of the stack structure **148** may also include remainders (e.g., remaining portions, unremoved portions) of the first insulative structures **104** (FIG. 1J) horizontally neighboring the conductive structures **142**.

(70) The first portions (e.g., the first portions **124-1A**, **124-2A**, **124-3A**, **124-4A** shown in FIG. 1J) of the dielectric liner structures **124** (e.g., the dielectric liner structures **124-1**, **124-2**, **124-3**, **124-4** shown in FIG. 1J) may be physically separated from the second portions (e.g., the second portions **124-1B**, **124-2B**, **124-3B**, **124-4B** shown in FIG. 1J) of the dielectric liner structures **124** (FIG. 1J) by removing (e.g., etching) regions of the dielectric liner structures **124** at intersections of the first portions and the second portions thereof, and then filling the resulting openings with at least one electrically insulative material having different etch selectivity than the dielectric liner structures **124** and the first insulative structures **104** (FIG. 1J) of the preliminary stack structure **102** (FIG. 1J). The electrically insulative material may, for example, have etch selectively substantially similar to that of the second insulative structures **106** (FIG. 1J) of preliminary stack structure **102** (FIG. 1J) and the dielectric spacer structures **122**. The material composition of the electrically insulative material may be substantially the same as or may be different than material composition(s) of the second insulative structures **106** (FIG. 1J) and the dielectric spacer structures **122** (FIG. 1J). In some embodiments, the electrically insulative material is formed of and includes SiO.sub.x (e.g., SiO.sub.2).

(71) In some embodiments, the material removal process employed to physically separate the first portions (e.g., the first portions **124-1A**, **124-2A**, **124-3A**, **124-4A** shown in FIG. 1J) and the second portions (e.g., the second portions **124-1B**, **124-2B**, **124-3B**, **124-4B** shown in FIG. 1J) of individual dielectric liner structures **124** (FIG. 1J) is controlled to substantially only physically separate the first portions and the second portions of the dielectric liner structures **124** from one another. The material removal process may, for example, form discrete openings vertically extending through individual dielectric liner structures **124**, but not through the dielectric spacer structures **122** horizontally neighboring the dielectric liner structures **124**. The discrete openings may then be filled with electrically insulative material to form discrete pillar structures (e.g., discrete circular column structures) horizontally interposed between the separated portions of the dielectric liner structures **124** in a first horizontal direction, and horizontally interposed between horizontally neighboring dielectric spacer structures **122** in a second horizontal direction orthogonal to the first horizontal direction.

(72) In additional embodiments, the material removal process employed to physically separate the first portions (e.g., the first portions **124-1A**, **124-2A**, **124-3A**, **124-4A** shown in FIG. 1J) and the second portions (e.g., the second portions **124-1B**, **124-2B**, **124-3B**, **124-4B** shown in FIG. 1J) of individual dielectric liner structures **124** (FIG. 1J) is controlled to also physically separate first portions (e.g., the first portions **122-1A**, **122-2A**, **122-3A**, **122-4A** shown in FIG. 1J) and second portions (e.g., the second portions **122-1B**, **122-2B**, **122-3B**, **122-4B** shown in FIG. 1J) of individual dielectric spacer structures **122** (e.g., the dielectric spacer structures **122-1**, **122-2**, **122-3**, **122-4** shown in FIG. 1J) from one another. The material removal process may, for example, form at least one linear opening (e.g., at least one trench) vertically extending through individual dielectric liner structures **124** as well as individual dielectric spacer structures **122** horizontally neighboring the individual dielectric liner structures **124**. The linear opening may then be filled with electrically insulative material to form at least one linear pillar structure (described in further detail below with

reference to FIG. 2) (e.g., a horizontally elongate, rectangular column structure) horizontally interposed between the separated portions of the individual dielectric liner structures **124** and the separated portions of the individual dielectric spacer structures **122** in a first horizontal direction; and continuously horizontally extending, in a second horizontal direction orthogonal to the first horizontal direction, through dielectric liner structures **124** and dielectric spacer structures **122** horizontally neighboring one another.

(73) In addition to physically separating at least the first portions (e.g., the first portions **124-1A**, **124-2A**, **124-3A**, **124-4A** shown in FIG. 1J) and the second portions (e.g., the second portions **124-1B**, **124-2B**, **124-3B**, **124-4B** shown in FIG. 1J) of individual dielectric liner structures **124** (FIG. 1J), structure(s) (e.g., pillar structures, linear pillar structure(s)) formed between the separated portions of individual dielectric liner structures **124** may provide structural support to the preliminary stack structure **102** (FIG. 1J) during so called “replacement gate” or “gate last” processing acts to form the conductive structures **142**, the conductive contact structures **144**, and the additional conductive contact structures **146**. The structure(s) may, for example, impede undesirable tier deformations (e.g., tier warping) and/or tier collapse during the formation of the conductive structures **142**, the conductive contact structures **144**, and the additional conductive contact structures **146**. One or more additional support structures (e.g., pillar structures, such as dielectric pillar structure) may be formed at desired locations within the preliminary stack structure **102** (FIG. 1J) prior to the replacement gate processing acts to provide further structural support to the preliminary stack structure **102** (FIG. 1J) during the formation of the conductive structures **142**, the conductive contact structures **144**, and the additional conductive contact structures **146**.

(74) With continued reference to FIG. 1K, quantities and positions (e.g., horizontal positions, vertical positions) of the conductive contact structures **144** may correspond to the quantities and positions of the first portions of the dielectric liner structures **124** (FIG. 1J) used to form the conductive contact structures **144**; and the quantities and positions (e.g., horizontal positions, vertical positions) of the additional conductive contact structures **146** may correspond to the quantities and positions of the second portions of the dielectric liner structures **124** (FIG. 1J) used to form the additional conductive contact structures **146**. As a non-limiting example, as shown in FIG. 1K, the conductive contact structures **144** may include a conductive contact structure **144-1**, a second conductive contact structure **144-2**, a third conductive contact structure **144-3**, and a fourth conductive contact structure **144-4** respectively corresponding to the first portion **124-1A** (FIG. 1J) of the first dielectric liner structure **124-1** (FIG. 1J), the first portion **124-2A** (FIG. 1J) of the second dielectric liner structure **124-2** (FIG. 1J), the first portion **124-3A** (FIG. 1J) of the third dielectric liner structure **124-3** (FIG. 1J), and the first portion **124-4A** (FIG. 1J) of the fourth dielectric liner structure **124-4** (FIG. 1J). As another non-limiting example, as also shown in FIG. 1K, the additional conductive contact structures **146** may include a first additional conductive contact structure **146-1**, a second additional conductive contact structure **146-2**, a third additional conductive contact structure **146-3**, and a fourth additional conductive contact structure **146-4** respectively corresponding to the second portion **124-1B** (FIG. 1J) of the first dielectric liner structure **124-1** (FIG. 1J), the second portion **124-2B** (FIG. 1J) of the second dielectric liner structure **124-2** (FIG. 1J), the second portion **124-3B** (FIG. 1J) of the third dielectric liner structure **124-3** (FIG. 1J), and the second portion **124-4B** (FIG. 1J) of the fourth dielectric liner structure **124-4** (FIG. 1J).

(75) As shown in FIG. 1K, the conductive contact structures **144** may be integral and continuous with some of the conductive structures **142** of the modified stack structure **148**, and the additional conductive contact structures **146** may be integral and continuous with some other of the conductive structures **142** of the modified stack structure **148**. For example, individual conductive contact structures **144** may be integral and continuous with individual conductive structures **142** of the modified stack structure **148** at the additional steps **134** of the reverse staircase structure **130** of the stadium structure **126**, and individual additional conductive contact structures **146** may be

integral and continuous with other individual conductive structures **142** of the modified stack structure **148** at the steps **132** of the forward staircase structure **128** of the stadium structure **126**. Individual conductive contact structures **144** and individual conductive structures **142** in physical contact therewith may constitute different portions of individual monolithic structures. As a non-limiting example, the first conductive contact structure **144-1** and one of the conductive structures **142** of the modified stack structure **148** in physical contact therewith (e.g., at an uppermost additional step **134** of the reverse staircase structure **130**) may together form a first monolithic structure comprising the first conductive contact structure **144-1** and the one of the conductive structures **142**. In addition, individual additional conductive contact structures **146** and other individual conductive structures **142** in physical contact therewith may constitute different portions of other individual monolithic structures. As a non-limiting example, the first additional conductive contact structure **146-1** and another of the conductive structures **142** of the modified stack structure **148** in physical contact therewith (e.g., at an uppermost step **132** of the forward staircase structure **128**) may together form a first additional monolithic structure comprising the first additional conductive contact structure **146-1** and the another of the conductive structures **142**.

(76) The conductive structures **142**, the conductive contact structures **144**, and the additional conductive contact structures **146** may each individually be formed of and include at least one electrically conductive material, such as one or more of a metal, an alloy, a conductive metal oxide, a conductive metal nitride, a conductive metal silicide, and a conductively doped semiconductor material. By way of non-limiting example, the conductive structures **142**, the conductive structures **142**, the conductive contact structures **144**, and the additional conductive contact structures **146** may each individually be formed of and include one or more of tungsten (W), tungsten nitride (WN.sub.y), nickel (Ni), tantalum (Ta), tantalum nitride (TaN.sub.y), tantalum silicide (TaSi.sub.x), platinum (Pt), copper (Cu), silver (Ag), gold (Au), aluminum (Al), molybdenum (Mo), titanium (Ti), titanium nitride (TiN.sub.y), titanium silicide (TiSi.sub.x), titanium silicon nitride (TiSi.sub.xN.sub.y), titanium aluminum nitride (TiAl.sub.xN.sub.y), molybdenum nitride (MoN.sub.x), iridium (Ir), iridium oxide (IrO.sub.z), ruthenium (Ru), ruthenium oxide (RuO.sub.z), and conductively doped silicon. In some embodiments, the conductive structures **142**, the conductive contact structures **144**, and the additional conductive contact structures **146** are each formed of and include tungsten (W).

(77) With continued reference to FIG. 1K, to form the modified stack structure **148** (including the conductive structures **142** and the insulative structures **150** thereof), the conductive contact structures **144**, and the additional conductive contact structures **146**, slots (e.g., slits, trenches) may be formed to vertically extend through the dielectric spacer structures **122**, the dielectric liner structures **124** (FIG. 1J), and the preliminary stack structure **102** (FIG. 1J) to form discrete blocks. The slots and the blocks may each extend in a first horizontal direction (e.g., the X-direction); and the slots and the blocks may alternate with one another in a second horizontal direction (e.g., the Y-direction) orthogonal to the first horizontal directions. The dielectric liner structures **124** (FIG. 1J) and the first insulative structures **104** (FIG. 1J) of the preliminary stack structure **102** (FIG. 1J) may then be selectively removed (e.g., selectively etched and exhumed), in whole or in part, through the slots. Thereafter, open volumes (e.g., void spaces) formed by the selective removal of the dielectric liner structures **124** (FIG. 1J) and the first insulative structures **104** (FIG. 1J) may be filled with electrically conductive material (e.g., tungsten) to form the conductive structures **142**, the conductive contact structures **144**, and the additional conductive contact structures **146**. The slots may then be filled with at least one dielectric material.

(78) Referring next to FIG. 1L, a dielectric structure **154** may be formed on or over the capping material **140**; portions of the dielectric structure **154** and the capping material **140** may be removed (e.g., etched) to form openings (e.g., vias, apertures) exposing portions of the conductive contact structures **144** and the additional conductive contact structures **146**; and then conductive plug structures **156** may be formed within the openings and in contact (e.g., physical contact, electrical

contact) with the conductive contact structures **144** and the additional conductive contact structures **146**. The conductive plug structures **156** may facilitate electrical communication between additional conductive structures (e.g., conductive line structures, such as access line structures) to be formed over the stadium structure **126** (and, hence, additional structures and/or devices to be electrically coupled to the additional conductive structures, such as one or more control devices) and the conductive contact structures **144** and the additional conductive contact structures **146**. (79) The dielectric structure **154** may be formed of and include at least one dielectric material having different etch selectivity than the capping material **140**. The dielectric structure **154** may serve as a mask to pattern the capping material **140**. The dielectric structure **154** may be selectively etchable relative to the capping material **140** during mutual exposure to a first etchant, and the capping material **140** may be selectively etchable relative to the dielectric structure **154** during mutual exposure to a second, different etchant. A material composition of the dielectric structure **154** is different than a material composition of the capping material **140**, and may comprise one or more of at least one dielectric oxide material (e.g., one or more of SiO₂, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO_x, HfO_x, NbO_x, TiO_x, ZrO_x, TaO_x, and MgO_x), at least one dielectric nitride material (e.g., SiN_y), at least one dielectric oxynitride material (e.g., SiO_xN_y), and at least one dielectric carboxynitride material (e.g., SiO_xC_zN_y). In some embodiments, the dielectric structure **154** is formed of and includes SiN_y (e.g., Si₃N₄). The dielectric structure **154** may be substantially homogeneous, or may be substantially heterogeneous. In some embodiments, the dielectric structure **154** is substantially homogeneous. In additional embodiments, the dielectric structure **154** is substantially heterogeneous. The dielectric structure **154** may be substantially planar, and may exhibit a desired thickness (e.g., height in the Z-direction).

(80) The conductive plug structures **156** may be formed of and include at least one electrically conductive material, such as one or more of a metal (e.g., W, Ti, Mo, Nb, V, Hf, Ta, Cr, Zr, Fe, Ru, Os, Co, Rh, Ir, Ni, Pd, Pt, Cu, Ag, Au, Al), an alloy (e.g., a Co-based alloy, an Fe-based alloy, an Ni-based alloy, an Fe- and Ni-based alloy, a Co- and Ni-based alloy, an Fe- and Co-based alloy, a Co- and Ni- and Fe-based alloy, an Al-based alloy, a Cu-based alloy, a Mg-based alloy, a Ti-based alloy, a steel, a low-carbon steel, a stainless steel), a conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide), and a conductively doped semiconductor material (e.g., conductively doped Si, conductively doped Ge, conductively doped SiGe). Each of the conductive plug structures **156** may have substantially the same material composition, or at least one of the conductive plug structures **156** may have a different material composition than at least one other of the conductive plug structures **156**.

(81) The conductive plug structures **156** may each individually be provided at a desired horizontal location (e.g., in the X-direction and the Y-direction) on or over one of the conductive contact structures **144** or one of the additional conductive contact structures **146**. The conductive plug structures **156** may each individually contact a single (e.g., only one) conductive contact structure **144** or a single (e.g., only one) additional conductive contact structure **146**. In some embodiments, each of the conductive plug structures **156** is individually substantially horizontally centered on one of the conductive contact structures **144** or one of the additional conductive contact structures **146**. In additional embodiments, one or more of the conductive plug structures **156** are individually horizontally offset (e.g., in the X-direction and/or in the Y-direction) from a horizontal center of the conductive contact structure **144** or the additional conductive contact structure **146** in contact therewith. In addition, as shown in FIG. 1L, in some embodiments, at least some of the conductive plug structures **156** are substantially horizontally aligned with one another. For example, at least some (e.g., all) conductive plug structures **156** horizontally neighboring one another in a first horizontal direction (e.g., the X-direction) may be substantially aligned with one another in a

second horizontal direction (e.g., the Y-direction) orthogonal to the first horizontal direction. In additional embodiments, at least some of the conductive plug structures **156** are offset from one another in multiple horizontal directions (e.g., the X-direction and the Y-direction). For example, at least some (e.g., all) conductive plug structures **156** horizontally neighboring one another in a first horizontal direction (e.g., the X-direction) may be offset from one another in a second horizontal direction (e.g., the Y-direction) orthogonal to the first horizontal direction.

(82) The dielectric structure **154** and the conductive plug structures **156** may be formed using conventional processes (e.g., conventional material deposition processes, conventional photolithography processes, conventional material removal processes) and conventional processing equipment, which are not described in detail herein.

(83) Thus, in accordance with embodiments of the disclosure, a method of forming a microelectronic device comprises forming a preliminary stack structure comprising a vertically alternating sequence of insulative structures and additional insulative structures arranged in tiers. Each of the tiers comprises one of the insulative structures and one of the additional insulative structures. A trench is formed to vertically extend into the preliminary stack structure. A horizontally alternating sequence of dielectric spacer structures and dielectric liner structures is formed within horizontal boundaries of the trench. At least some of the dielectric liner structures physically contact different insulative structures of the preliminary stack structure than one another. Dielectric material of the dielectric liner structures and the insulative structures of the preliminary stack structure is at least partially replaced with electrically conductive material.

(84) Furthermore, in accordance with embodiments of the disclosure, a microelectronic device comprises a stack structure, a stadium structure within the stack structure, and conductive contact structures. The stack structure comprises a vertically alternating sequence of conductive structures and insulative structures arranged in tiers. Each of the tiers comprises one of the conductive structures and one of the insulative structures. The stadium structure comprises a forward staircase structure having steps comprising edges of the tiers, and a reverse staircase structure opposing the forward staircase structure and having additional steps comprising additional edges of the tiers. The conductive contact structures vertically extend to upper vertical boundaries of at least some of the conductive structures of the stack structure at the steps of the forward staircase structure and the additional steps of the reverse staircase structure. The conductive contact structures are each integral and continuous with one of the conductive structures.

(85) Microelectronic device structures (e.g., the microelectronic device structure **100** previously described with reference to FIG. **1L**) in accordance with embodiments of the disclosure may be included in embodiments of microelectronic devices (e.g., memory devices, such as a 3D NAND Flash memory devices) of the disclosure. For example, FIG. **2** illustrates a simplified, partial top-down view of a microelectronic device **200** including a microelectronic device structure **201**. The microelectronic device structure **201** may be substantially similar to the microelectronic device structure **100** at the processing stage previously described with reference to FIG. **1L**.

(86) Referring to FIG. **2**, the microelectronic device **200** includes a stack structure **202** including a vertically alternating (e.g., in the Z-direction) sequence of conductive structures and insulative structures arranged in tiers. The stack structure **202** may be substantially similar to the modified stack structure **148** previously described with reference to FIG. **1K**. As shown in FIG. **2**, the stack structure **202** may be partitioned in the Y-direction by slots **203**. The slots **203** may be filled with dielectric material, and correspond to the slots previously described with reference to FIG. **1K** in relation to the formation of the conductive structures **142**, the conductive contact structures **144**, and the additional conductive contact structures **146** through so called “replacement gate” or “gate last” processing acts. The slots **203** may divide (e.g., in the Y-direction) the stack structure **202** into multiple blocks **205**. In addition, the stack structure **202** includes a memory array region **207**, and a distributed stadium region **209** horizontally neighboring (e.g., in the X-direction) the memory array region **207**. As described in further detail below, the microelectronic device **200** further includes

additional components (e.g., features, structures, devices) within boundaries of the different horizontal regions (e.g., the memory array region **207**, the distributed stadium region **209**) of the stack structure **202**.

(87) Within horizontal boundaries of the memory array region **207** of the stack structure **202**, the microelectronic device **200** may include vertically extending pillar structures **211**. The vertically extending pillar structures **211** may, for example, vertically extend (e.g., in the Z-direction) from or proximate data lines **215** (e.g., digit lines, bit lines), through the memory array region **207** of stack structure **202**, and to or proximate a source structure **217** (e.g., a source plate). In FIG. 2, the source structure **217** is represented with dashed lines to indicate that the source structure **217** may vertically underlie the stack structure **202**. Each of the vertically extending pillar structures **211** may include a semiconductive pillar (e.g., a polysilicon pillar, a silicon-germanium pillar) at least partially surrounded by one or more charge storage structures (e.g., a charge trapping structure, such as a charge trapping structure comprising an oxide-nitride-oxide (“ONO”) material; floating gate structures). Intersections of the vertically extending pillar structures **211** and conductive structures (e.g., corresponding to the conductive structures **142** previously described with reference to FIG. 1K) of tiers (e.g., corresponding to the modified tiers **152** previously described with reference to FIG. 1K) of the stack structure **202** may define vertically extending (e.g., in the Z-direction) strings of memory cells **223** coupled in series with one another within the memory array region **207** of the stack structure **202**. In some embodiments, the memory cells **223** formed at the intersections of the conductive structures of the tiers of the stack structure **202** and the vertically extending pillar structures comprise so-called “MONOS” (metal—oxide—nitride—oxide—semiconductor) memory cells. In additional embodiments, the memory cells **223** comprise so-called “TANOS” (tantalum nitride—aluminum oxide—nitride—oxide—semiconductor) memory cells, or so-called “BETANOS” (band/barrier engineered TANOS) memory cells, each of which are subsets of MONOS memory cells. In further embodiments, the memory cells **223** comprise so-called “floating gate” memory cells including floating gates (e.g., metallic floating gates) as charge storage structures. The floating gates may horizontally intervene between the central structures of the vertically extending pillar structures **211** and the conductive structures of the different tiers of the stack structure **202**. The microelectronic device **200** may include any desired quantity and distribution of the vertically extending pillar structures **211** within the memory array region **207** of the stack structure **202**.

(88) With continued reference to FIG. 2, within horizontal boundaries of the distributed stadium region **209** of the stack structure **202**, the microelectronic device **200** may include stadium structures **226** distributed within the stack structure **202**. The stadium structures **226** may individually have a geometric configuration (e.g., a shape and dimensions) substantially similar to that previously described with respect to the stadium structure **126** (including geometric configurations of the forward staircase structure **128**, the reverse staircase structure **130**, and the central region **136** thereof) (FIGS. 1H through 1L), except at least some of the stadium structures **226** may be positioned at different elevations (e.g., vertical positions in the Z-direction) than one another within the stack structure **202**. Aside from at least some differing elevations within the stack structure **202**, each of the stadium structures **226** may have substantially the same geometric configuration (e.g., substantially the same shape and size, including substantially the same shapes and sizes of individual components thereof, and substantially the same shapes and sizes resulting from the combination of different components thereof) as each other of the stadium structures **226**. In additional embodiment, at least one of the stadium structures **226** has a different geometric configuration (e.g., a different shape and/or one or more different dimensions, including substantially a different shape and/or one or more different dimensions of one or more individual components thereof, and/or a different shape and/or one or more different dimensions resulting from the combination of different components thereof) than at least one other of the stadium structures **226**.

(89) As shown in FIG. 2, portions of individual blocks **205** of the stack structure **202** within the distributed stadium region **209** of the stack structure **202** may include sections of individual stadium structures **226** separated (e.g., divided, segmented, isolated) from other sections of the individual stadium structures **226** by the slots **203**. For example, a first section **226A** of an individual (e.g., single, one) stadium structure **226** may be located within and horizontally extend (e.g., in the Y-direction) partially (e.g., less than completely) across a first of the blocks **205** of the stack structure **202**, and a second section **226B** of the individual stadium structure **226** may be located within and horizontally extend (e.g., in the Y-direction) partially (e.g., less than completely) across a second of the blocks **205** horizontally neighboring (e.g., in the Y-direction) the first of the blocks **205**. One of the slots **203** may be horizontally interposed between (e.g., in the Y-direction) the first section **226A** of the individual stadium structure **226** and the second section **226B** of the individual stadium structure **226**. Individual blocks **205** of the stack structure **202** may include first sections **226A** of multiple of the stadium structures **226**, and second sections **226B** of multiple other of the stadium structures **226**. As shown in FIG. 2, within an individual block **205** of the stack structure **202**, the first sections **226A** of multiple of the stadium structures **226** may be substantially horizontally aligned with one another in a first horizontal direction (e.g., the X-direction), the second sections **226B** of multiple other of the stadium structures **226** may also be substantially horizontally aligned with one another in the first horizontal direction (e.g., the X-direction), and the first sections **226A** of the multiple of the stadium structures **226** may horizontally neighbor the second sections **226B** of the multiple other of the stadium structures **226** in a second horizontal direction (e.g., the Y-direction) orthogonal to the first horizontal direction.

(90) With continued reference to FIG. 2, within individual blocks **205** of the stack structure **202**, a word line bridging region **213** may continuously horizontally extend from and between sections of different stadium structures **226** located within horizontal boundaries (e.g., in the X-direction and the Y-direction) of the block **205** of the stack structure **202**. As shown in FIG. 2, for a given block **205** of the stack structure **202**, the word line bridging region **213** thereof may continuously horizontally extend from and between the first sections **226A** and the second sections **226B** of different stadium structures **226** located within the horizontal boundaries of the block **205**. The word line bridging regions **213** may include portions of the conductive structures (e.g., corresponding to the conductive structures **142** previously described with reference to FIG. 1K) of the stack structure **202** continuously horizontally extending from and between the sections of the different stadium structures **226**, and may be established during the so-called “replacement gate” or “gate last” processing acts used to form the conductive structures of the stack structure **202**. The word line bridging regions **213** facilitate appropriate word line driving using center-of-die staircase positions and configurations.

(91) Still referring to FIG. 2, within individual blocks **205** of the stack structure **202**, conductive contact structures **244** and additional conductive contact structure **246** may be integral and continuous with conductive structures of the stack structure **202** at steps (e.g., corresponding to the steps **132** previously described with reference to FIGS. 1H through 1L) and additional steps (e.g., corresponding to the additional steps **134** previously described with reference to FIGS. 1H through 1L) of sections (e.g., the first sections **226A**, the second sections **226B**) of the stadium structures **226**. The conductive contact structures **244** and the additional conductive contact structure **246** may correspond to the conductive contact structures **144** and the additional conductive contact structure **146** previously described with reference to FIG. 1K. As shown in FIG. 2, dielectric spacer structures **222** may horizontally intervene between and separate horizontally neighboring conductive contact structures **244** and horizontally neighboring additional conductive contact structures **246**; and additional dielectric structures **245** may horizontally intervene between and separate conductive contact structures **244** horizontally neighboring additional conductive contact structures **246**. The dielectric spacer structures **222** may correspond to the dielectric spacer structures **122** previously described with reference to FIGS. 1I through 1L; and the additional

dielectric structures **245** may correspond to the discrete pillar structures and/or the linear pillar structures previously described with reference to FIG. **1K**.

(92) In addition, within individual blocks **205** of the stack structure **202**, conductive plug structures **256** may contact (e.g., physically contact, electrically contact) the conductive contact structures **244** and the additional conductive contact structure **246**. The conductive plug structures **256** may correspond to the conductive plug structures **156** previously described with reference to FIG. **1L**. The conductive plug structures **256** may, for example, be connected (e.g., physically connected, electrically connected) to access lines **219** (e.g., word lines) of the microelectronic device **200**. The access lines **219** may be formed of and include at least one electrically conductive material, such as one or more of a metal (e.g., W, Ti, Mo, Nb, V, Hf, Ta, Cr, Zr, Fe, Ru, Os, Co, Rh, Ir, Ni, Pd, Pt, Cu, Ag, Au, Al), an alloy (e.g., a Co-based alloy, an Fe-based alloy, an Ni-based alloy, an Fe- and Ni-based alloy, a Co- and Ni-based alloy, an Fe- and Co-based alloy, a Co- and Ni- and Fe-based alloy, an Al-based alloy, a Cu-based alloy, a Mg-based alloy, a Ti-based alloy, a steel, a low-carbon steel, a stainless steel), a conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide), and a conductively doped semiconductor material (e.g., conductively doped Si, conductively doped Ge, conductively doped SiGe).

(93) Still referring to FIG. **2**, the microelectronic device **200** may further include at least one control unit **221** (e.g., a control device). The control unit **221** may include one or more of string driver circuitry, pass gates, circuitry for selecting gates, circuitry for selecting conductive lines, circuitry for amplifying signals, and circuitry for sensing signals. The control unit **221** may, for example, be electrically coupled to the data lines **215**, the source structure **217**, and the access lines **219**. In some embodiments, the control unit **221** vertically underlies (e.g., in the Z-direction) and is substantially confined within horizontal boundaries of the memory array region **207** of the stack structure **202**. In FIG. **2**, the control unit **221** is represented with dashed lines to indicate that the control unit **221** may vertically underlie the stack structure **202**. The control unit **221** may vertically underlie the source structure **217** of the microelectronic device **200**. In some embodiments, the control unit **221** includes CMOS circuitry. In some such embodiments, the control unit **221** may be characterized as having a “CMOS under Array” (“CuA”) configuration.

(94) Referring next to FIG. **3**, microelectronic device structures (e.g., the microelectronic device structure **100** previously described with reference to FIG. **1L**) in accordance with embodiments of the disclosure may be included in additional embodiments of microelectronic devices (e.g., additional memory devices, such as additional 3D NAND Flash memory devices) of the disclosure. For example, FIG. **3** illustrates a simplified, partial top-down view of a microelectronic device **300** including a microelectronic device structure **301**. The microelectronic device structure **301** may be substantially similar to the microelectronic device structure **100** at the processing stage previously described with reference to FIG. **1L**. Throughout FIG. **3** and the associated description below, features (e.g., structures, materials, regions) functionally similar to features of the microelectronic device **200** previously described with reference to FIG. **2** are referred to with similar reference numerals incremented by 100. To avoid repetition, not all features shown in FIG. **3** are described in detail herein. Rather, unless described otherwise below, in FIG. **3**, a feature designated by a reference numeral that is a 100 increment of the reference numeral of a feature previously described with reference to FIG. **2** will be understood to be substantially similar to the previously described feature.

(95) As shown in FIG. **3**, within horizontal boundaries of a distributed stadium region **309** of the stack structure **302**, the microelectronic device **300** may include stadium structures **326** having different horizontal geometric configurations (e.g., different horizontal shapes, different horizontal dimensions) than the stadium structures **226** of the microelectronic device **200** previously described with reference to FIG. **2**. For example, individual stadium structures **326** within the distributed stadium region **309** may be configured such that different sections of an individual stadium

structure **326** horizontally extend substantially (e.g., completely) across individual blocks **305** of the stack structure **302** in a first horizontal direction (e.g., the Y-direction) and are separated from one another by the slots **303** horizontally extending a second horizontal direction (e.g., the X-direction) orthogonal to the first horizontal direction. For example, a first section **326A** of an individual (e.g., single, one) stadium structure **326** may be located within and substantially horizontally extend across (e.g., in the Y-direction) a first of the blocks **305** of the stack structure **302**, and a second section **326B** of the individual stadium structure **226** may be located within and substantially horizontally extend across (e.g., in the Y-direction) a second of the blocks **205** horizontally neighboring (e.g., in the Y-direction) the first of the blocks **205**. One of the slots **303** may be horizontally interposed between (e.g., in the Y-direction) the first section **326A** of the individual stadium structure **326** and the second section **326B** of the individual stadium structure **326**. A first block **305** of the stack structure **302** may include first sections **326A** of multiple of the stadium structures **326** separated from one another in the X-direction and each substantially horizontally extending across the first block **305** in the Y-direction; and a second block **305** horizontally neighboring the first block **305** in the Y-direction may include second sections **326B** of the multiple of the stadium structures **326** separated from one another in the X-direction and each substantially horizontally extending across the second block **305** in the Y-direction.

(96) As shown in FIG. 3, given the different horizontal geometric configurations of the stadium structures **326** of the microelectronic device **300** relative to those of the stadium structures **226** (FIG. 2) of the microelectronic device **200** (FIG. 2), individual blocks **305** of the stack structure **302** are free of word line bridging regions substantially similar to the word line bridging regions **213** previously described with reference to FIG. 2. Rather, as different sections (e.g., the first section **326A**, the second section **326B**) of individual stadium structures **326** substantially horizontally extend (e.g., in the Y-direction) across the individual blocks **305** of the stack structure **302** associated therewith, such word line bridging regions may be absent (e.g., omitted) from the blocks **305** of the stack structure **302**. The configuration of the microelectronic device **300** shown in FIG. 3 facilitates appropriate word line driving using end-of-die staircase positions and configurations.

(97) Thus, in accordance with embodiments of the disclosure, a memory device comprises a stack structure, stadium structures, conductive contact structures, and strings of memory cells. The stack structure has tiers each comprising a conductive structure and an insulative structure vertically neighboring the conductive structure. The stadium structures are positioned at different vertical depths within the stack structure than one another. Each of the stadium structures comprises a forward staircase structure having steps comprising horizontal ends of a group of the tiers, and a reverse staircase structure opposing the forward staircase structure and having additional steps comprising additional horizontal ends of the group of the tiers. The conductive contact structures vertically extend from at least some of the conductive structures of the stack structure at the steps and the additional steps of the stadium structures. The conductive contact structures are each integral and continuous with one of the conductive structures. The strings of memory cells vertical extend through the stack structure.

(98) With returned reference to FIG. 1L, in additional embodiments, a microelectronic device structure of the disclosure is formed to have a different configuration than the microelectronic device structure **100** at the processing stage depicted in FIG. 1L. By way of non-limiting example, FIGS. 4A through 4C are simplified partial cross-sectional views illustrating a method of forming a microelectronic device structure **400**, in accordance with additional embodiments of the disclosure. The method of forming the microelectronic device structure **400** up to the processing stage depicted in FIG. 4A incorporates the processing acts and features previously described in relation to the formation of the microelectronic device structure **100** up to and including the processing stage previously described with reference to FIG. 1E. The processing stages depicted in FIGS. 4A through 4C and described in further detail below may, for example, include processing act

performed in place of and/or in combination with the processing acts previously described with reference to FIGS. **1F** through **1L** to form the microelectronic device structure **100** previously described with reference to FIG. **1L**. Throughout FIGS. **4A** through **4C** and the associated description below, features (e.g., structures, materials, regions) functionally similar to features of the microelectronic device structure **100** previously described with reference to one or more of FIGS. **1A** through **1L** are referred to with similar reference numerals incremented by 100. To avoid repetition, not all features shown in FIGS. **4A** through **4C** are described in detail herein. Rather, unless described otherwise below, in FIGS. **4A** through **4C**, a feature designated by a reference numeral that is a 100 increment of the reference numeral of a feature previously described with reference to one or more of FIGS. **1A** through **1L** will be understood to be substantially similar to the previously described feature.

(99) As shown in FIG. **4A**, following the formation of the first dielectric liner structure **424-1** (e.g., through processing acts substantially similar to those previously described with reference to FIG. **1E**), the microelectronic device structure **400** may be subjected to at least one additional material removal (e.g., etching, such as anisotropic dry etching) processes to increase the vertical depths (e.g., in the Z-direction) of remaining (e.g., unfilled) portions of the trench **416** formed in the preliminary stack structure **402**. As described in further detail below, the additional material removal process may increase the vertical depths of the remaining portions of the trench **416** by relatively greater magnitudes than the increases to the vertical depths of the remaining portions of the trench **116** previously described with reference to FIG. **1F**. The additional material removal process may increase a vertical depth of a remaining portion of the first region **416A** of the trench **416**, and may also increase a vertical depth of a remaining portion of the second region **416B** of the trench **416**. Following the additional material removal process, the remaining portion of the second region **416B** of the trench **416** may vertically extend to a relatively lower depth within the preliminary stack structure **402** than the remaining portion of the first region **416A** of the trench **416**. A magnitude of a vertical offset between a lower vertical boundary of the first region **416A** of the trench **416** and a lower vertical boundary of the second region **416B** of the trench **416** may be substantially maintained at the end of the additional material removal process. For example, the additional material removal process may increase a vertical depth of each of the remaining portion of the first region **416A** of the trench **416** and the remaining portion of the second region **416B** of the trench **416** by a distance (e.g., in the Z-direction) greater than or equal to the combined height (e.g., in the Z-direction) of at least two (e.g., at least three) tiers **408** of the preliminary stack structure **402**. As shown in FIG. **4A**, in some embodiments, the additional material removal process increases a vertical depth of each of the remaining portion of the first region **416A** of the trench **416** and the remaining portion of the second region **416B** of the trench **416** by a distance substantially equal the combined height of three (3) of the tiers **408** of the preliminary stack structure **402**.

(100) Referring next to FIG. **4B**, following the processing stage described with reference to FIG. **4A**, the microelectronic device structure **400** may be subjected to additional processing acts substantially similar to those previously described with reference to the FIGS. **1D**, **1E**, and **4A** to fill remaining portions of the trench **416** (FIG. **4A**) with more of the dielectric liner structures **424** (e.g., in addition to the first dielectric liner structure **424-1**) and more of the dielectric spacer structures **422** (e.g., in addition to the first dielectric spacer structure **422-1**), and form a stadium structure **426** vertically underlying and in contact with the dielectric liner structures **424** and the dielectric spacer structures **422**. As shown in FIG. **4B**, the stadium structure **426** may include a forward staircase structure **428** including steps **432** each individually comprising horizontal ends of multiple (e.g., greater than or equal to three (3), greater than or equal to four (4)) of the tiers **408** of the preliminary stack structure **402**; a reverse staircase structure **430** including additional steps **434** each individually comprising horizontal ends of multiple (e.g., greater than or equal to three (3), greater than or equal to four (4)) of the tiers **408**; and a central region **436** horizontally interposed between the forward staircase structure **428** and the reverse staircase structure **430**. The central

region **436** of the stadium structure **426** may, for example, constitute an intersection region and vertical termination location for the forward staircase structure **428** and the reverse staircase structure **430**. Different dielectric spacer structures **422** and different dielectric liner structures **424** may be formed to physically contact different steps **432** of the forward staircase structure **428** and different additional steps **434** of the reverse staircase structure **430** than one another, in substantially the same manner previously described for the dielectric spacer structures **122** and different dielectric liner structures **124** with reference to FIG. **1H**.

(101) As shown in FIG. **4B**, in some embodiments, individual steps **432** of the forward staircase structure **428** of the stadium structure **426** are formed to comprise horizontal ends of four (4) of the tiers **408** of the preliminary stack structure **402**; and individual additional steps **434** of the reverse staircase structure **430** of the stadium structure **426** are also formed to comprise horizontal ends of four (4) of the tiers **408** of the preliminary stack structure **402**. Individual steps **432** of the stadium structure **426** vertically neighboring individual additional steps **434** of the stadium structure **426** may share multiple (e.g., three (3)) tiers **408** of the preliminary stack structure **402** with the individual additional steps **434**. For example, three (3) of the four (4) tiers **408** of the preliminary stack structure **402** associated with an individual step **432** may also be three (3) of the four (4) tiers **408** of the preliminary stack structure **402** associated with an individual additional step **434** vertically neighboring the individual step **432**. One of the three (3) tiers **408** shared by an individual step **432** and an individual additional step **434** vertically neighboring the individual step **432** may form a vertically upper portion of the individual step **432** and a vertically lower portion of the individual additional step **434**, or vice versa. As a non-limiting example, an uppermost additional step **434** of the reverse staircase structure **430** may comprise horizontal ends of a first group of four (4) tiers **408** of the preliminary stack structure **402**; and an uppermost step **432** of the forward staircase structure **428** may comprise horizontal ends of a second group of four (4) tiers **408** of the preliminary stack structure **102**, wherein three (3) of the four (4) tiers **408** of the second group are the same as three (3) of the four (4) tiers **408** of the first group. A relatively highest tier **408** of the first group of four (4) tiers **408** may form an upper portion of the uppermost additional step **434** and may physically contact the first portion **424-1A** of the first dielectric liner structure **424-1**; and three (3) relatively lower tiers **408** of the first group of four (4) tiers **408** may form a lower portion of the uppermost additional step **434** and may not physically contact the first portion **424-1A** of the first dielectric liner structure **424-1**. In turn, a relatively highest tier **408** of the three (3) relatively lower tiers **408** forming the lower portion of the upper additional step **434** may constitute a relatively highest tier **408** of the second group of four (4) tiers **408** and may form an upper portion of the uppermost step **432** in physical contact with the second portion **424-1B** of the first dielectric liner structure **424-1**; and three (3) relatively lower tiers **408** of the second group of four (4) tiers **408** may form a lower portion of the uppermost step **432** and may not physically contact the second portion **424-1B** of the first dielectric liner structure **424-1**.

(102) Referring next to FIG. **4C**, following the processing stage depicted in FIG. **4B**, the microelectronic device structure **400** may be subjected to additional processing acts substantially similar to those previously described with reference to FIGS. **1I** through **1L** to attain the configuration of the microelectronic device structure **400** depicted in FIG. **4C**. Accordingly, the method of forming the microelectronic device structure **400** following the processing stage depicted in FIG. **4B** incorporates the processing stages and features previously described in relation to the formation of the microelectronic device structure **100** from the processing stage previously described with reference to FIG. **1I** through the processing stage previously described with reference to FIG. **1L**.

(103) The microelectronic device structure **400** described with reference to FIG. **4C** may be included in embodiments of microelectronic devices (e.g., memory devices, such as a 3D NAND Flash memory devices) of the disclosure. As a non-limiting example, the microelectronic device structure **400** at the processing stage depicted in FIG. **4C** may be included in the microelectronic

device **200** previously described with reference to FIG. 2 as the microelectronic device structure **201** (FIG. 2). In some such embodiments, the stadium structures **226** (FIG. 2) may individually have a geometric configuration (e.g., a shape and dimensions) substantially similar to that previously described with respect to the stadium structure **426** (including geometric configurations of the forward staircase structure **428**, the reverse staircase structure **430**, and the central region **436** thereof) (FIGS. 4B and 4C), except at least some of the stadium structures **226** (FIG. 2) may be positioned at different elevations (e.g., vertical positions in the Z-direction) than one another within the stack structure **202**. In addition, within individual blocks **205** (FIG. 2) of the stack structure **202** (FIG. 2), the conductive contact structures **244** (FIG. 2), the additional conductive contact structures **246** (FIG. 2), and the dielectric spacer structures **222** (FIG. 2) may respectively correspond to the conductive contact structures **444**, the additional conductive contact structure **446**, and the dielectric spacer structures **422** of the microelectronic device structure **400** at the processing stage depicted in FIG. 4C. Within an individual block **205** (FIG. 2) of the stack structure **202** (FIG. 2), a first section **226A** (FIG. 2) of an individual (e.g., single, one) stadium structure **226** (FIG. 2) may include steps **432** and additional steps **434** that are vertically offset (e.g., vertically above, vertically below) from steps **432** and additional steps **434** of a second section **226B** (FIG. 2) of an additional individual stadium structure **226** (FIG. 2). The steps **432** of the forward staircase structure **428** of the first section **226A** (FIG. 2) of the stadium structure **226** (FIG. 2) may be vertically offset (e.g., vertically above or vertically below), by one (1) tier **408**, from the steps **432** of the forward staircase structure **428** of the second section **226B** (FIG. 2) of the additional stadium structure **226** (FIG. 2); and the additional steps **434** of the reverse staircase structure **440** of the first section **226A** (FIG. 2) of the stadium structure **226** (FIG. 2) may be vertically offset (e.g., vertically above or vertically below), by one (1) tier **408**, from the additional steps **434** of the reverse staircase structure **430** of the second section **226B** (FIG. 2) of the additional stadium structure **226** (FIG. 2).

(104) The additional steps **434** of the first section **226A** of the stadium structure **226** may each individually be vertically offset from vertically neighboring steps **432** and additional steps **434** of the second section **226B** of the additional stadium structure **226**.

(105) In further embodiments, a microelectronic device structure of the disclosure is formed to have a different configuration than each of the microelectronic device structure **100** at the processing stage depicted in FIG. 1L and the microelectronic device structure **400** at the processing stage depicted in FIG. 4C. By way of non-limiting example, FIGS. 5A through 5G are simplified partial cross-sectional views illustrating a method of forming a microelectronic device structure **500**, in accordance with additional embodiments of the disclosure. The processing stages depicted in FIGS. 5A through 5G and described in further detail below may, for example, include processing act performed in place of and/or in combination with the processing acts previously described with reference to FIGS. 1A through 1L to form the microelectronic device structure **100** previously described with reference to FIG. 1L. Throughout FIGS. 5A through 5G and the associated description below, features (e.g., structures, materials, regions) functionally similar to features of the microelectronic device structure **100** previously described with reference to one or more of FIGS. 1A through 1L are referred to with similar reference numerals incremented by 100. To avoid repetition, not all features shown in FIGS. 5A through 5G are described in detail herein. Rather, unless described otherwise below, in FIGS. 5A through 5G, a feature designated by a reference numeral that is a 100 increment of the reference numeral of a feature previously described with reference to one or more of FIGS. 1A through 1L will be understood to be substantially similar to the previously described feature.

(106) Referring to FIG. 5A, a microelectronic device structure **500** may be formed to include a preliminary stack structure **502**, and a patterned mask structure **510** on or over the preliminary stack structure **502**. The preliminary stack structure **502** includes a vertically alternating (e.g., in the Z-direction) sequence of first insulative structures **504** and second insulative structures **506**

arranged in tiers **508**. The patterned mask structure **510** may include at least one opening **512** (e.g., aperture, via) vertically extending therethrough. The preliminary stack structure **502** (including the first insulative structures **504** and the second insulative structures **506** thereof) and the patterned mask structure **510** (including the opening **512** therein) may be formed to respectively be substantially similar to the preliminary stack structure **102** (including the first insulative structures **104** and the second insulative structures **106** thereof) and the patterned mask structure **110** (including the opening **112** therein) previously described herein with reference to FIG. **1A**. (107) Referring next to FIG. **5B**, the microelectronic device structure **500** may be subjected to one or more material removal processes (e.g., one or more chopping processes) to form at least one trench **516** (e.g., opening, blind via) vertically extending (e.g., in the Z-direction) into the preliminary stack structure **502**. As shown in FIG. **5B**, the trench **516** within the preliminary stack structure **502** may be substantially confined within horizontal boundaries of the opening **512** within the patterned mask structure **510**. In embodiments wherein the patterned mask structure **510** includes multiple openings **512**, different trenches **516** may be formed to extend to different vertical depths than one another within the preliminary stack structure **502**. For example, at least one of the trenches **516** may vertically extend to a relatively lower depth within the preliminary stack structure **502** than at least one other of the trenches **516**. The vertical depths of the trenches **516** relative to one another may at least partially depend on the quantity of tiers **508** of the preliminary stack structure **502**, the quantity of the trenches **516** within the preliminary stack structure **502**, and the horizontal dimensions of the trenches **516**. The trench(es) **516** may be configured to facilitate subsequent formation of vertically extending insulative structures in physical contact with at least some (e.g., each) of the tiers **508** of the preliminary stack structure **502**, as described in further detail below.

(108) As shown in FIG. **5B**, the trench **516** may be formed to include a lower vertical boundary **518** (e.g., floor) and horizontal boundaries **520** (e.g., sides). The lower vertical boundary **518** of the trench **516** may be formed to be substantially horizontally planar. For example, prior to forming the trench **516**, the processing stage previously described with reference to FIG. **1B** (in relation to the formation of the microelectronic device structure **100**) may be omitted. Accordingly, different horizontal regions of the trench **516** may all vertically extend (e.g., in the Z-direction) to substantially the same vertical elevation within the preliminary stack structure **502**. In addition, as depicted in FIG. **5B**, the horizontal boundaries **520** of the trench **516** may be substantially vertically planar.

(109) The trench **516** in the preliminary stack structure **502** may be formed to vertically terminate (e.g., vertically end) at the second insulative structure **506** of a single (e.g., only one) tier **508** of the preliminary stack structure **502**. In additional embodiments, the trench **516** in the preliminary stack structure **502** may be formed to vertically terminate at the first insulative structures **504** of a single (e.g., only one) tier **508** of the preliminary stack structure **502**.

(110) Referring next to FIG. **5C**, a first dielectric spacer structure **522-1** may be formed within the opening **512** in the patterned mask structure **510** and the trench **516** in the preliminary stack structure **502**. The first dielectric spacer structure **522-1** may partially (e.g., less than completely) fill each of the opening **512** and the trench **516**, and may be formed on exposed surfaces of the patterned mask structure **510** and the preliminary stack structure **502** within horizontal boundaries (e.g., in the X-direction and the Y-direction) of the trench **516** (and, hence, the opening **512** in the patterned mask structure **510**). For example, as shown in FIG. **5C**, the first dielectric spacer structure **522-1** may be formed directly horizontally adjacent (e.g., horizontally on) side surfaces of the patterned mask structure **510** and the preliminary stack structure **502** at (e.g., coplanar with) the horizontal boundaries **520** of the trench **516**, and may also be formed directly vertically adjacent (e.g., vertically on) an upper surface of the tier **508** of the preliminary stack structure **502** defining the lower vertical boundary **518** (FIG. **5B**) of the trench **516**. The first dielectric spacer structure **522-1** may substantially vertically extend across (e.g., in the Z-direction) and cover the side

surfaces of the patterned mask structure **510** and the preliminary stack structure **502** at (e.g., coplanar with) the horizontal boundaries **520** of the trench **516**, and may only partially horizontally extend across (e.g., in the X-direction and the Y-direction) and cover the upper surface of the tier **508** of the preliminary stack structure **502** defining the lower vertical boundary **518** (FIG. 5B) of the trench **516**. As shown in FIG. 5C, the first dielectric spacer structure **522-1** may include a first portion **522-1A** and a second portion **522-1B**. The first portion **522-1A** and the second portion **522-1B** may vertically extend (e.g., in the Z-direction) to substantially the same depth within the preliminary stack structure **502** as one another.

(111) The first dielectric spacer structure **522-1** may have substantially the same horizontal dimensions (e.g., width) and material composition as the first dielectric spacer structure **122-1** previously described with reference to FIG. 1D. In addition, the first dielectric spacer structure **522-1** may be formed in substantially the same manner as the first dielectric spacer structure **122-1** previously described with reference to FIG. 1D.

(112) Referring next to FIG. 5D, a first dielectric liner structure **524-1** may be formed within remaining (e.g., unfilled) portions of the opening **512** in the patterned mask structure **510** and the trench **516** in the preliminary stack structure **502**. The first dielectric liner structure **524-1** may partially (e.g., less than completely) fill the remaining portions of each of the opening **512** and the trench **516**, and may be formed on exposed surfaces of the first dielectric spacer structure **522-1** (FIG. 5D) and the preliminary stack structure **502** within horizontal boundaries (e.g., in the X-direction and the Y-direction) of the trench **516** (and, hence, the opening **512** in the patterned mask structure **510**). For example, the first dielectric liner structure **524-1** may be formed directly horizontally adjacent (e.g., horizontally on) inner side surfaces of the first dielectric spacer structure **522-1**, and may also be formed directly vertically adjacent (e.g., vertically on) an upper surface of a first insulative structure **504** of a tier **508** of the preliminary stack structure **502** at a modified lower vertical boundary of the trench **516** formed in the processing stage previously described with reference to FIG. 5C. The first dielectric liner structure **524-1** may substantially vertically extend across (e.g., in the Z-direction) and cover the inner side surfaces of the first dielectric spacer structure **522-1**.

(113) As shown in FIG. 5D, the first dielectric liner structure **524-1** may include a first portion **524-1A** and a second portion **524-1B**. The first portion **524-1A** may be formed directly horizontally adjacent (e.g., horizontally on) the first portion **522-1A** of the first dielectric spacer structure **522-1**, and the second portion **524-1B** may be formed directly horizontally adjacent (e.g., horizontally on) the second portion **522-1B** of the first dielectric spacer structure **522-1**. The first portion **524-1A** and the second portion **524-1B** may vertically extend (e.g., in the Z-direction) to substantially the same depth within the preliminary stack structure **502** as one another. The vertical depth of each of the first portion **524-1A** and the second portion **524-1B** of the first dielectric liner structure **524-1** may be greater than or equal to the vertical depth of the first dielectric spacer structure **522-1** (including the first portion **522-1A** and the second portion **522-1B** thereof). For example, the first portion **524-1A** and the second portion **524-1B** of the first dielectric liner structure **524-1** may each vertically extend to and terminate at the first insulative structure **504** of a tier **508** of the preliminary stack structure **502** vertically neighboring a lower vertical boundary of the first dielectric spacer structure **522-1**.

(114) The first dielectric liner structure **524-1** may have substantially the same horizontal dimensions (e.g., width) and material composition as the first dielectric liner structure **124-1** previously described with reference to FIG. 1E. In addition, the first dielectric liner structure **524-1** may be formed in substantially the same manner as the first dielectric liner structure **124-1** previously described with reference to FIG. 1E.

(115) Referring next to FIG. 5E, a second dielectric spacer structure **522-2** may be formed within a remaining (e.g., unfilled) portion the opening **512** in the patterned mask structure **510** and in remaining, deepened portions of the trench **516** in the preliminary stack structure **502**. The second

dielectric spacer structure **522-2** may partially (e.g., less than completely) fill the remaining portion of the opening **512** and the remaining, deepened portions of the trench **516**; and may be formed on exposed surfaces of the first dielectric liner structure **524-1** and the preliminary stack structure **502** within horizontal boundaries (e.g., in the X-direction and the Y-direction) of the trench **516** (and, hence, the opening **512** in the patterned mask structure **510**). For example, the second dielectric spacer structure **522-2** may be formed directly horizontally adjacent (e.g., horizontally on) exposed inner side surfaces of the first dielectric liner structure **524-1** and the preliminary stack structure **502** within the trench **516**. The second dielectric spacer structure **522-2** and may also be formed directly vertically adjacent (e.g., vertically on) upper surfaces of a second insulative structure **506** of a tier **508** of the preliminary stack structure **502** at a modified lower vertical boundary of the trench **516** formed in the processing stage previously described with reference to FIG. 5D. The second dielectric spacer structure **522-2** may substantially vertically extend across (e.g., in the Z-direction) and cover the inner side surfaces of the first dielectric liner structure **524-1**.

(116) As shown in FIG. 5E, the second dielectric spacer structure **522-2** may include a first portion **522-2A** and a second portion **522-2B**. The first portion **522-2A** of the second dielectric spacer structure **522-2** may be formed directly horizontally adjacent (e.g., horizontally on) the first portion **524-1A** of the first dielectric liner structure **524-1**, and the second portion **522-2B** may be formed directly horizontally adjacent (e.g., horizontally on) the second portion **524-1B** of the first dielectric liner structure **524-1**. The first portion **522-2A** and the second portion **522-2B** may vertically extend (e.g., in the Z-direction) to substantially the same depth within the preliminary stack structure **502** as one another. The vertical depth of each of the first portion **522-2A** and the second portion **522-2B** of the second dielectric spacer structure **522-2** may be greater than or equal to the vertical depth of the first dielectric liner structure **524-1** (including the first portion **524-1A** and the second portion **524-1B** thereof). For example, the first portion **522-2A** and the second portion **522-2B** of the second dielectric spacer structure **522-2** may each vertically extend to and terminate at the second insulative structure **506** of a tier **508** of the preliminary stack structure **502** vertically neighboring a lower vertical boundary of the first dielectric liner structure **524-1**.

(117) The second dielectric spacer structure **522-2** may have substantially the same horizontal dimensions (e.g., width) and material composition as the second dielectric spacer structure **122-2** previously described with reference to FIG. 1G. In addition, the second dielectric spacer structure **522-2** may be formed in substantially the same manner as the second dielectric spacer structure **122-2** previously described with reference to FIG. 1G.

(118) Referring next to FIG. 5F, following the processing stage described with reference to FIG. 5E, the microelectronic device structure **500** may be subjected to additional processing acts similar to those previously described with reference to FIGS. 5D and 5E to fill remaining portions of the trench **516** (FIG. 5E) with more of the dielectric liner structures **524** and more of the dielectric spacer structures **522**, and form a stadium structure **526** vertically underlying and in contact with the dielectric liner structures **524** and the dielectric spacer structures **522**. As shown in FIG. 5F, the stadium structure **526** may include a forward staircase structure **528** including steps **532** comprising horizontal ends of tiers **508** of the preliminary stack structure **502**; a reverse staircase structure **530** including additional steps **534** comprising additional horizontal ends of the tiers **508** of the preliminary stack structure **502**; and a central region **536** horizontally interposed between the forward staircase structure **528** and the reverse staircase structure **530**. The central region **536** of the stadium structure **526** may, for example, constitute an intersection region and vertical termination location for the forward staircase structure **528** and the reverse staircase structure **530**. As described in further detail below, different dielectric spacer structures **522** and different dielectric liner structures **524** may be formed to physically contact different steps **532** of the forward staircase structure **528** and different additional steps **534** of the reverse staircase structure **530** than one another.

(119) As shown in FIG. 5F, the dielectric liner structures **524** may physically contact the first

insulative structures **504** of different tiers **508** of the preliminary stack structure **502** at the steps **532** and the additional steps **534** of the stadium structure **526**. As shown in FIG. 5F, first portions **524-1A**, **524-2A**, **524-3A**, **524-4A** of different dielectric liner structures **524** (e.g., the first dielectric liner structure **524-1**, a second dielectric liner structure **524-2**, a third dielectric liner structure **524-3**, a fourth dielectric liner structure **524-4**) may physically contact at least some of the additional steps **534** of the reverse staircase structure **530** of the stadium structure **526**; and second portions **524-1B**, **524-2B**, **524-3B**, **524-4B** of the different dielectric liner structures **524** (e.g., the first dielectric liner structure **524-1**, the second dielectric liner structure **524-2**, the third dielectric liner structure **524-3**, the fourth dielectric liner structure **524-4**) may physically contact at least some of the steps **532** of the forward staircase structure **528** of the stadium structure **526**. The dielectric liner structures **524** may have horizontal positions and horizontal dimensions substantially similar to those of the dielectric liner structures **124** previously described with reference to FIG. 1H.

(120) The dielectric spacer structures **522** may be formed horizontally adjacent the dielectric spacer structures **522** over different steps **532** and additional steps **534** of the stadium structure **526**. As shown in FIG. 5F, first portions **522-1A**, **522-2A**, **522-3A**, **522-4A** of different dielectric spacer structures **522** (e.g., the first dielectric spacer structure **522-1**, the second dielectric spacer structure **522-2**, a third dielectric spacer structure **522-3**, a fourth dielectric spacer structure **522-4**) may be formed over and within horizontal boundaries of at least some of the additional steps **534** of the reverse staircase structure **530** of the stadium structure **526**; and second portions **522-1B**, **522-2B**, **522-3B**, **522-4B** of the different dielectric spacer structures **522** (e.g., the first dielectric spacer structure **522-1**, the second dielectric spacer structure **522-2**, the third dielectric spacer structure **522-3**, the fourth dielectric spacer structure **522-4**) may be formed over and within horizontal boundaries of at least some of the steps **532** of the forward staircase structure **528** of the stadium structure **526**. The fifth dielectric spacer structure **522-5** may be formed over and within horizontal boundaries of the central region **536** of the stadium structure **526**. The dielectric spacer structures **522** may have horizontal positions and horizontal dimensions substantially similar to those of the dielectric spacer structures **122** previously described with reference to FIG. 1H.

(121) Still referring to FIG. 5F, the steps **532** of the forward staircase structure **528** of the stadium structure **526** may each individually comprise horizontal ends of one (1) tier **508** of the preliminary stack structure **502**. In addition, the additional steps **534** of the reverse staircase structure **530** of the stadium structure **526** may also each individually comprise horizontal ends of one (1) tier **508** of the preliminary stack structure **502**. Individual steps **532** and individual additional steps **534** of the stadium structure **526** at substantially the same elevation (e.g., in the Z-direction) within the preliminary stack structure **502** may share a tier **508** of the preliminary stack structure **502** with one another. For example, an individual step **532** (e.g., an uppermost step **532**) of the forward staircase structure **528** may comprise first horizontal ends of an individual tier **508** of the preliminary stack structure **502**; and an individual additional step **534** (e.g., an uppermost additional step **534**) of the reverse staircase structure **530** may be located at substantially the same vertical position as the individual step **532** of the forward staircase structure **528** and may comprise second horizontal ends of the individual tier **508** of the preliminary stack structure **502**.

(122) Referring next to FIG. 5G, following the processing stage depicted in FIG. 5F, the microelectronic device structure **500** may be subjected to additional processing acts substantially similar to those previously described with reference to FIGS. 1I through 1L to attain the configuration of the microelectronic device structure **500** depicted in FIG. 5G. Accordingly, the method of forming the microelectronic device structure **500** following the processing stage depicted in FIG. 5F incorporates the processing stages and features previously described in relation to the formation of the microelectronic device structure **100** from the processing stage previously described with reference to FIG. 1I through the processing stage previously described with reference to FIG. 1L.

(123) The microelectronic device structure **500** described with reference to FIG. 5G may be included in embodiments of microelectronic devices (e.g., memory devices, such as a 3D NAND Flash memory devices) of the disclosure. By way of non-limiting example, the microelectronic device structure **500** at the processing stage depicted in FIG. 5G may be included in the microelectronic device **200** (FIG. 2) as the microelectronic device structure **201** (FIG. 2), or may be included in microelectronic device **300** (FIG. 3) as the microelectronic device structure **301** (FIG. 3). In some such embodiments, the stadium structures **226** (FIG. 2) or the stadium structures **326** (FIG. 3) may individually have a geometric configuration (e.g., a shape and dimensions) substantially similar to that previously described with respect to the stadium structure **526** (including geometric configurations of the forward staircase structure **528**, the reverse staircase structure **530**, and the central region **536** thereof) (FIGS. 5F and 5G), except at least some of the stadium structures **226** (FIG. 2) or at least some of the stadium structures **326** (FIG. 3) may be positioned at different elevations (e.g., vertical positions in the Z-direction) than one another within the stack structure **202** (FIG. 2) or the stack structure **302** (FIG. 3). In addition, within individual blocks **205** (FIG. 2) of the stack structure **202** (FIG. 2) or individual blocks **305** (FIG. 3) of the stack structure **302** (FIG. 3), the conductive contact structures **244** (FIG. 2) or the conductive contact structures **344** (FIG. 3), the additional conductive contact structures **246** (FIG. 2) or the additional conductive contact structures **346** (FIG. 3), and the dielectric spacer structures **222** (FIG. 2) or the dielectric spacer structures **322** (FIG. 3) may respectively correspond to the conductive contact structures **544**, the additional conductive contact structure **546**, and the dielectric spacer structures **522** of the microelectronic device structure **500** at the processing stage depicted in FIG. 5G.

(124) Microelectronic device structures (e.g., the microelectronic device structure **100** previously described with reference to FIG. 1L; the microelectronic device structure **400** previously described with reference to FIG. 4C; the microelectronic device structure **500** previously described with reference to FIG. 5G) and microelectronic devices (e.g., the microelectronic device **200** previously described with reference to FIG. 2; the microelectronic device **300** previously described with reference to FIG. 3) in accordance with embodiments of the disclosure may be used in embodiments of electronic systems of the disclosure. For example, FIG. 6 is a block diagram of an illustrative electronic system **600** according to embodiments of disclosure. The electronic system **600** may comprise, for example, a computer or computer hardware component, a server or other networking hardware component, a cellular telephone, a digital camera, a personal digital assistant (PDA), portable media (e.g., music) player, a Wi-Fi or cellular-enabled tablet such as, for example, an iPad® or SURFACE® tablet, an electronic book, a navigation device, etc. The electronic system **600** includes at least one memory device **602**. The memory device **602** may comprise, for example, an embodiment of one or more of a microelectronic device structure and a microelectronic device previously described herein. The electronic system **600** may further include at least one electronic signal processor device **604** (often referred to as a “microprocessor”). The electronic signal processor device **604** may, optionally, include an embodiment of one or more of a microelectronic device structure and a microelectronic device previously described herein. While the memory device **602** and the electronic signal processor device **604** are depicted as two (2) separate devices in FIG. 6, in additional embodiments, a single (e.g., only one) memory/processor device having the functionalities of the memory device **602** and the electronic signal processor device **604** is included in the electronic system **600**. In such embodiments, the memory/processor device may include one or more of a microelectronic device structure and a microelectronic device previously described herein. The electronic system **600** may further include one or more input devices **606** for inputting information into the electronic system **600** by a user, such as, for example, a mouse or other pointing device, a keyboard, a touchpad, a button, or a control panel. The electronic system **600** may further include one or more output devices **608** for outputting information (e.g., visual or audio output) to a user such as, for example, one or more of a monitor, a display, a printer, an audio

output jack, and a speaker. In some embodiments, the input device **606** and the output device **608** may comprise a single touchscreen device that can be used both to input information to the electronic system **600** and to output visual information to a user. The input device **606** and the output device **608** may communicate electrically with one or more of the memory device **602** and the electronic signal processor device **604**.

(125) Thus, in accordance with embodiments of the disclosure, an electronic system comprises an input device, an output device, a processor device operably coupled to the input device and the output device, and a memory device operably coupled to the processor device. The memory device comprises a microelectronic device structure comprising a stack structure, a stadium structure, and conductive contact structures. The stack structure comprises a vertically alternating sequence of conductive structures and insulative structures arranged in tiers. The stadium structure is within the stack structure and comprises opposing staircase structures each having steps comprising edges of at least some of the tiers. The conductive contact structures are integral and continuous with at least some of the conductive structures of the stack structure at the steps of the opposing staircase structures of the stadium structure. Each of the conductive contact structures individually has an inner horizontal boundary substantially coplanar with an inner horizontal boundary of one of the steps of the opposing staircase structures.

(126) The methods, structures (e.g., the microelectronic device structures **100**, **400**, **500**), devices (e.g., the microelectronic devices **200**, **300**), and systems (e.g., the electronic system **600**) of the disclosure advantageously facilitate one or more of improved performance, reliability, and durability, lower costs, increased miniaturization of components, improved pattern quality, and greater packaging density as compared to conventional structures, conventional devices, and conventional systems. The methods and structures of the disclosure may alleviate problems related to the formation and processing of conventional microelectronic devices including stack structures having staircase structures at edges thereof. For example, the methods and structures of the disclosure do not suffer from the relatively small sizing and spacing error margins associated with properly forming staircase structures of conventional microelectronic device structures to receive contact structures thereon. In addition, the methods and structures of the disclosure may reduce the risk of undesirable damage (e.g., contact punch through) as well as undesirable current leakage and short circuits as compared to conventional methods and conventional structures.

(127) While the disclosure is susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and have been described in detail herein. However, the disclosure is not limited to the particular forms disclosed. Rather, the disclosure is to cover all modifications, equivalents, and alternatives falling within the scope of the following appended claims and their legal equivalents.

Claims

1. A microelectronic device, comprising: blocks horizontally extending in parallel a first direction and respectively comprising: tiers each including conductive material and insulative material vertically neighboring the conductive material; a staircase structure having steps comprising edges of the tiers; and conductive contacts integral and continuous with the conductive material of the tiers at the steps of the staircase structure, the conductive contacts horizontally positioned at terminal ends of the steps in the first direction; dielectric spacer structures on the steps of the staircase structure and respectively horizontally extending from and between two of the conductive contacts horizontally neighboring one another in the first direction; and slot structures horizontally extending in parallel in the first direction and horizontally alternating with the blocks in a second direction orthogonal to the first direction.
2. The microelectronic device of claim 1, wherein the conductive contacts respectively have a non-circular horizontal cross-sectional shape.

3. The microelectronic device of claim 2, wherein the conductive contacts respectively have a rectangular horizontal cross-sectional shape, with a largest horizontal dimension thereof extending in the second direction.
4. The microelectronic device of claim 2, further comprising conductive plugs on upper surfaces of the conductive contacts, the conductive plugs having different horizontal cross-sectional shapes and different horizontal cross-sectional areas than the conductive contacts.
5. The microelectronic device of claim 1, further comprising pillar structures respectively comprising semiconductor material within horizontal areas of and vertically extending through the blocks.
6. A microelectronic device, comprising: blocks horizontally extending in parallel a first direction and respectively comprising: tiers each including conductive material and insulative material vertically neighboring the conductive material; a staircase structure having steps comprising edges of the tiers; and conductive contacts integral and continuous with the conductive material of the tiers at the steps of the staircase structure, the conductive contacts respectively having a non-circular horizontal cross-sectional shape and respectively comprising: a first portion substantially linearly extending in the first direction; a second portion substantially linearly extending in a second direction; and a third portion interposed between the first portion and the second portion and horizontally extending in an at least partially arcuate path; and slot structures horizontally extending in parallel in the first direction and horizontally alternating with the blocks in the second direction orthogonal to the first direction.
7. A memory device, comprising: blocks horizontally extending in parallel a first direction and each comprising tiers individually including conductive material and insulative material vertically neighboring the conductive material, the blocks respectively comprising: a memory array region; and a distributed stadium region neighboring the memory array region in the first direction and comprising: stadium structures each including staircase structures respectively having steps comprising horizontal ends of some of the tiers; and conductive contacts within horizontal areas of the stadium structures and individually integral and continuous with the conductive material of a respective one of the tiers, the conductive contacts respectively having a rectangular horizontal cross-sectional shape, with a largest horizontal dimension of the rectangular horizontal cross-sectional shape extending in a second direction; and vertically extending strings of memory cells individually within a horizontal area of the memory array region of respective ones of the blocks.
8. The memory device of claim 7, wherein the stadium structures respectively comprise: a first staircase structure having positive slope; and a second staircase structure having negative slope, treads of the steps of the second staircase structure located at different vertical positions than other treads of the steps of the first staircase structure.
9. The memory device of claim 7, wherein the conductive contacts are respectively horizontally offset, in the first direction, from a horizontal center of a respective one of the steps of one of the stadium structures most vertically proximate thereto.
10. The memory device of claim 7, further comprising dielectric oxide spacers within horizontal areas of the stadium structures, the dielectric oxide spacers individually horizontally extending, in at least the first direction, from a side surface of one of the conductive contacts to and an additional side surface of an additional one of the conductive contacts.
11. The memory device of claim 7, further comprising filled slots horizontally extending in parallel in the first direction and horizontally alternating with the blocks in a second direction perpendicular to the first direction.
12. The memory device of claim 11, wherein, for a respective one of the blocks: a first group of the stadium structures is horizontally adjacent one of the filled slots in the second direction; and a second group of the stadium structures is horizontally adjacent an additional one of the filled slots in the second direction, at least some of the stadium structures of the second group of the stadium structures horizontally overlapping at least some of the stadium structures of the first group of the

stadium structures in the first direction.

13. The memory device of claim 11, wherein, for a respective one of the blocks, the stadium structures individually horizontally extend, in the second direction, from one of the filled slots to an additional one of the filled slots.

14. A 3D NAND Flash memory device, comprising: two blocks horizontally extending in parallel in a first direction, each of the two blocks comprising: tiers respectively including conductive material and insulative material vertically neighboring the conductive material; stadium structures respectively including: a staircase structure having steps comprising horizontal ends of some of the tiers; and an additional staircase structure opposing the staircase structure in the first direction and having additional steps comprising horizontal ends of some other of the tiers; and conductive contacts within horizontal areas of the stadium structures and individually integral and continuous with the conductive material of a respective one of the tiers, the conductive contacts respectively having a non-circular horizontal cross-sectional shape; and conductive plugs on upper surfaces of the conductive contacts, the conductive plugs having different horizontal cross-sectional shapes and different horizontal cross-sectional areas than the conductive contacts; a slot structure horizontally interposed between the two blocks in a second direction orthogonal to the first direction; and strings of memory cells within horizontal areas of and vertically extending through the two blocks.

15. The 3D NAND Flash memory device of claim 14, wherein for respective ones of the stadium structures, treads of the steps of the staircase structure thereof are all vertically offset from additional treads of the additional steps of the additional staircase structure thereof.

16. The 3D NAND Flash memory device of claim 15, wherein for respective ones of the stadium structures: a group of the conductive contacts is within a horizontal area of the staircase structure; and an additional group of the conductive contacts is within an additional horizontal area of the additional staircase structure.

17. The 3D NAND Flash memory device of claim 14, wherein each of the two blocks respectively comprises two rows of the stadium structures horizontally extending in the first direction.
